First Order Differential Equation:

A differential equation is an equation which expresses an explicit or implicit relationship between a function $y=f(t)$ and one or more of its derivatives or differentials. Example of differential equation include

$$\frac{dy}{dt}=5t+9 \quad y'=12y \quad and$$

$$y''-2y+19=0$$

Equations involving a single independent variable, such as those above, are called ordinary differential equations. The solution or integral of a differential equation is any equation, without derivative or differential that is defined over an interval and satisfies the differential equation for all the values of the independent variables in the interval. The order of a differential equation is the order of the highest derivative in the equation. The degree of a differential equation is the highest power to which the derivative of highest order is raised.

First Order Linear Differential Equation with Constant Coefficient and Constant term.

Suppose we are seeking the solution of the first order linear differential equation.

$$\frac{dy}{dt}+ay=b$$

Where a and b are two constants. The general solution will consist of the sum of two components:

A particular integral y_p and complementary function

Complementary: y_c

Complementary function y_c is the general solution of the reduced equation or homogenous equation, i.e.

$$\frac{dy}{dt}+ay=0$$

Complementary function represents the deviation from equilibrium.

Let us try a simplest possible type of solution of the form

$$y=Ae^{rt} \rightarrow (i) \qquad (with \quad Ae^{rt}\neq 0)$$

Then

$$y'=rAe^{rt}$$

Substituting into above homogenous equation

$$rAe^{rt}+aAe^{rt}=0$$

$$Ae^{rt}(r+a)=0$$

Dividing both sides by Ae^{rt}, we get

$$r+a=0$$

$$r \quad =-a$$

Substituting into (i)

$$y_c=Ae^{-at} \qquad when \quad a\neq 0$$

Particular Integral: y_p

Particular integral is any solution of the complete non−homogenous equation, i.e.

$$\frac{dy}{dt}+ay=b$$

Particular integral represents the inter-temporal equilibrium level of y.

Try a simplest possible type of solution, i.e. y = k

This implies that $y'=0$. Hence above non−homogenous equation will become

$$aK=b$$

$$y=K=\frac{b}{a}$$

Hence

$$y_p=\frac{b}{a} \qquad where \quad a\neq 0$$

General solution:

$$y_{(t)}=y_c+y_p$$

$$y_{(t)}=Ae^{-at}+\frac{b}{a} \qquad (a\neq 0)$$

General solution

In order to definitize the solution set t =0

$$y_{(0)}=Ae^{0}+\frac{b}{a}$$

$$A=y_{(0)}-\frac{b}{a} \qquad (As \ \ e^{0}=1)$$

Substituting into above general solution

$$y_{(t)}=\left[y_{(0)}-\frac{b}{a}\right]e^{-at}+\frac{b}{a} \qquad (where \ \ a\neq 0)$$

Definite solution

In case, if a=0 then the differential equation will become

$$\frac{dy}{dt}=b$$

In this case the complementary function will

$$y_{C}=Ae^{-at}=Ae^{0}=A \qquad (a=0)$$

For particular integral, we will try a new solution of the form

$$y=Kt$$

Then

$$y'=K$$

Substituting into above non−homogenous equation

$$K=b$$

$$y_{p}=Kt=bt \qquad (a=0)$$

And general solution will be

$$y_{(t)}=y_{c}+y_{p}$$

$$y_{(t)}=A+bt \qquad (when \ \ a=0)$$

To definitize solution set t=0

$$y_{(0)}=A$$

Substituting into above

$$y_{(t)}=y_{(0)}+bt$$

Verification of the solution:

It is true of all solutions of differential equations that their validity can always be checked by differentiation.

Let us check the solution

$$y_{(t)}=\left[y_{(0)}-\frac{b}{a}\right]e^{-at}+\frac{b}{a} \qquad \rightarrow (a)$$

We can obtain the derivative

$$\frac{dy}{dt}=-a\left[y_{(0)}-\frac{b}{a}\right]e^{-at}$$

When both expressions $\frac{dy}{dt}$ and $y_{(t)}$ are substitutes into the left side of the original differential equation, that side should reduce exactly to the value of the constant term b on the right side of the equation if the solution is correct. By performing this substitution, we get

$$-a\left[y_{(0)}-\frac{b}{a}\right]e^{-at}+a\left\{\left[y_{(0)}-\frac{b}{a}\right]e^{-at}+\frac{b}{a}\right\}=b$$

Thus our solution is correct, provided it also satisfies the initial condition. To check the latter let us set t=0 in the solution “a”, the result will be

$y_{(0)}=\left[y_{(0)}-\frac{b}{a}\right]+\frac{b}{a}=y_{(0)}$, is an identity, the initial condition is indeed satisfied.

It is recommended that as a final step in the process of solving a differential equation, you make it a habit to check the validity of your answer by making sure that

1. The derivative of the time path $y_{(t)}$ is consistent with the given differential equation and
2. The definite solution satisfies the initial condition.

Example 1:

$$\frac{dy}{dt}+2y=6 \qquad y_{(0)}=10$$

$$a=2 \qquad b=6$$

Applying formula

$$y_{(t)}=\left[y_{(0)}-\frac{b}{a}\right]e^{-at}+\frac{b}{a}$$

$$y_{(t)}=\left[10-\frac{6}{2}\right]e^{-2t}+\frac{6}{2}$$

$$y_{(t)}=(10-3)\,e^{-2t}+3$$

$$y_{(t)}=7\,e^{-2t}+3$$

Example 2:

$$\frac{dy}{dt}+4y=0 \qquad y_{(0)}=1$$

Since $a=4 \qquad b=0$

$$y_{(t)}=\left[y_{(0)}-\frac{b}{a}\right]e^{-at}+\frac{b}{a}$$

$$y_{(t)}=[1-0]\,e^{-4t}+0$$

$$y_{(t)}=e^{-4t}$$

Example 3:

$$\frac{dy}{dt}=2 \qquad y_{(0)}=5$$

Since $a=0$

$$y_{(t)}=y_{(0)}+bt$$

$$y_{(t)}=5+2t$$

Exercise 14.1

1. Find y_c. y_p, the general solution and the definite solution, given:

(a) $\dfrac{dy}{dt}+4y=12 \qquad y_{(0)}=2$

$$\frac{dy}{dt}+4y=0$$

$$\frac{dy}{dt}=-4y$$

$$\frac{1}{y}\frac{dy}{dt}=-4$$

$$\frac{1}{y}\,dy=-4\,dt$$

$$\int\frac{1}{y}dy=-4\int dt$$

$$\ln y+c_1=-4t+c_2$$

$$\ln y=-4t+c \qquad c=c_2-c_1$$

Taking antilog

$$e^{\ln y}=e^{-4t+c}$$

$$y_C=Ae^{-4t} \qquad e^c=A$$

$$\frac{dy}{dt}+4y=12$$

$$Let \quad y=k \quad then \quad \frac{dy}{dt}=0$$

$$0+4k=12$$

$$k=12/4=3$$

$$y=k=3$$

$$y_p=3$$

$$y_{(t)}=y_c+y_p$$

$$y_{(t)}=Ae^{-4t}+3 \qquad General\ \ Solution$$

To definite set t=0

$$y_{(0)}=A+3$$

$$2=A+3$$

$$2-3=A$$

$$A=-1$$

$$y_{(t)}=-e^{-4t}+3 \qquad Definite\ \ Solution$$

(b) $\dfrac{dy}{dt}-2y=0 \qquad y_{(0)}=9$

$$\frac{dy}{dt}=2y$$

$$\frac{1}{y}dy=2\,dt$$

$$\int\frac{1}{y}dy=2\int dt$$

$$\ln y+c_1=2t+c_2$$

$$\ln y=2t+c \qquad c=c_2-c_1$$

Taking antilog

$$e^{\ln y}=e^{2t}\,.\,e^c$$

$$y_{(t)}=Ae^{2t} \qquad General\ \ Solution \quad e^c=A$$

To definitize set t=0

$$y_{(0)}=A$$

$$9=A$$

$$y_{(t)}=9\,e^{2t} \qquad Ddefinite\ \ Solution$$

(c) $\dfrac{dy}{dt}+10y=15 \qquad y_{(0)}=0$

$$\frac{dy}{dt}+10y=0$$

$$\frac{dy}{dt}=-10y$$

$$\frac{1}{y}dy=-10\,dt$$

$$\int\frac{1}{y}dy=-10\int dt$$

$$\ln y+c_1=-10t+c_2$$

$$\ln y=-10t+c \qquad c=c_2-c_1$$

Taking antilog

$$e^{\ln y}=e^{-10t}\,.\,e^c$$

$$y_C=Ae^{-10t}$$

Complementary function

$$\frac{dy}{dt}+10y=15$$

$Let\ \ y=k\ \ \ then\ \ \frac{dy}{dt}=0$

$10y=15$

$y=\frac{15}{10}=\frac{3}{2}$

$y_p=\frac{3}{2}$

$y_{(t)}=y_c+y_p$

$=Ae^{-10t}+\frac{3}{2}\qquad General\ \ Solution$

To definite solution set t=o

$y_{(0)}=A+\frac{3}{2}$

$A=-\frac{3}{2}\qquad As\ \ y_{(0)}=0$

Hence $y_{(t)}=-\frac{3}{2}e^{-10t}+\frac{3}{2}$

$y_{(t)}=\frac{3}{2}\left(1-e^{-10t}\right)$

(d) $2\frac{dy}{dt}+4y=6\qquad y_{(0)}=1\frac{1}{2}$

Dividing through by 2

$\frac{dy}{dt}+2y=3$

$\frac{dy}{dt}+2y=0$

$\frac{dy}{dt}=-2y$

$\frac{1}{y}dy=-2dt$

$\int\frac{1}{y}dy=-2\int dt$

$\ln y+c_1=-2t+c_2$

$\ln y=-2t+c\qquad c=c_2-c_1$

Taking antilog

$e^{\ln y}=e^{-2t}.\ e^c$

$y_c=Ae^{-2t}\qquad e^c=A$

$\frac{dy}{dt}+2y=3$

$Let\ \ \ y=K\ \ \ Then\ \ \frac{dy}{dt}=0$

$2y=3$

$y=\frac{3}{2}$

$y_p=\frac{3}{2}$

$y_{(t)}=y_c+y_p$

$y_{(t)}=Ae^{-2t}+\frac{3}{2}\qquad General\ \ Solution$

Set t=0

$y_{(0)}=A+\frac{3}{2}$

$\frac{3}{2}-\frac{3}{2}=A$

$A=0$

Hence $y_{(t)}=\frac{3}{2}\qquad Definite\ \ \ Solution$

Q. 2. Check the validity of your answer to the preceding problem.

(a) $y_{(t)}=-e^{-4t}+3$

Taking derivative W.r.t “t”

$\frac{dy}{dt}=4e^{-4t}$

Putting into the original equation

$\frac{dy}{dt}+4y=12$

$4e^{-4t}+4\left(-e^{-4t}+3\right)=12$

$4e^{-4t}-4e^{-4t}+12=12$

$12=12$

(b) $y_{(t)}=9e^{2t}$

Taking derivative W.r.t “t”

$\frac{dy}{dt}=18e^{2t}$

Putting into the original equation

$\frac{dy}{dt}-2y=0$

$18e^{2t}-2\left(9e^{2t}\right)=0$

$18e^{2t}-18e^{2t}=0$

$0=0$

(c) $y_{(t)}=\frac{3}{2}-\frac{3}{2}e^{-10t}$

$\frac{dy}{dt}=15e^{-10t}$

Substituting into original equation

$$\frac{dy}{dt}=+10y=15$$

$$15e^{-10t}+10\left(\frac{3}{2}-\frac{3}{2}e^{-10t}\right)=15$$

$$15e^{-10t}+15-15e^{-10t}=15$$

$$15=15$$

(d) $y_{(t)}=\frac{3}{2}$

$$\frac{dy}{dt}=0$$

Substituting into original equation

$$2\frac{dy}{dt}+4y=6$$

$$2(0)+4\left(\frac{3}{2}\right)=6$$

$$6=6$$

Q. 3. Find the solution of each of the following by using an appropriate formula developed in the text.

(a) $\frac{dy}{dt}+y=4 \qquad y_{(0)}=0$

$a=1 \qquad b=4$

$$y_{(t)}=\left[y_{(0)}-\frac{b}{a}\right]e^{-at}+\frac{b}{a}$$

$$y_{(t)}=\left(0-\frac{4}{1}\right)e^{-t}+\frac{4}{1}$$

$$y_{(t)}=-4e^{-t}+4$$

$$y_{(t)}=4(1-e^{-t})$$

(b) $\frac{dy}{dt}=23 \qquad y_{(0)}=1$

$a=0 \qquad b=23$

Applying formula

$$y_{(t)}=y_{(0)}+bt$$

$$y_{(t)}=1+23t$$

(c) $\frac{dy}{dt}-5y=0 \qquad y_{(o)}=6$

$a=-5, \qquad b=0$

$$y_{(t)}=y_{(0)}e^{-at}$$

$$y_{(t)}=6e^{-(-5t)}$$

$$y_{(t)}=6e^{5t}$$

(d) $\frac{dy}{dt}+3y=2 \qquad y_{(0)}=4$

$a=3, \qquad b=2$

$$y_{(t)}=\left(y_{(0)}-\frac{b}{a}\right)e^{-at}+\frac{b}{a}$$

$$y_{(t)}=\left(4-\frac{2}{3}\right)e^{-3t}+\frac{2}{3}$$

$$y_{(t)}=\frac{10}{3}e^{-3t}+\frac{2}{3}$$

(e) $\frac{dy}{dt}-7y=7 \qquad y_{(0)}=7$

$a=-7 \qquad b=7$

$$y_{(t)}=\left[y_{(0)}-\frac{b}{a}\right]e^{-at}+\frac{b}{a}$$

$$y_{(t)}=\left(7-\frac{7}{-7}\right)e^{7t}+\frac{7}{(-7)}$$

$$y_{(t)}=(7+1)e^{7t}-1$$

$$y_{(t)}=8e^{7t}-1$$

(f) $\frac{3dy}{dt}+6y=5 \qquad y_{(0)}=0$

Dividing by 3

$\frac{dy}{dt}+2y=\frac{5}{3} \qquad b=\frac{5}{3}$

$a=2$

$$y_{(t)}=\left[y_{(0)}-\frac{b}{a}\right]e^{-at}+\frac{b}{a}$$

$$y_{(t)}=\left(0-\frac{5/3}{2}\right)e^{-2t}+\frac{5/3}{2}$$

$$y_{(t)}=-\frac{5}{6}e^{-2t}+\frac{5}{6}$$

$$y_{(t)}=\frac{5}{6}(1-e^{-2t})$$

Q. 4. Check the validity of your answer to the preceding problem.

(a) $y_{(t)}=4(1-e^{-t})$

$$y_{(t)}=4-4e^{-t}$$

$$\frac{dy}{dt}=4e^{-t}$$

Substituting values in the original equation

$$\frac{dy}{dt}+y=4$$

$$4e^{-t}+4(1-e^{-t})=4$$
$$4e^{-t}+4-4e^{-t}=4$$
$$4=4$$

(b) $y_{(t)}=1+23t$

$$\frac{dy}{dt}=23$$

Substituting values in the original equation

$$\frac{dy}{dt}=23$$
$$23=23$$

(c) $y_{(t)}=6e^{5t}$

$$\frac{dy}{dt}=30e^{5t}$$

Substituting into the equation

$$\frac{dy}{dt}-5y=0$$
$$30e^{5t}-5(6e^{5t})=0$$
$$30e^{5t}-30e^{5t}=0$$
$$0=0$$

(d) $y_{(t)}=\frac{10}{3}e^{-3t}+\frac{2}{3}$

$$\frac{dy}{dt}=-10e^{-3t}$$

Substituting into given differential equation

$$\frac{dy}{dt}+3y=2$$
$$-10e^{-3t}+3\left(\frac{10}{3}e^{-3t}+\frac{2}{3}\right)=2$$
$$-10e^{-3t}+10e^{-3t}+2=2$$
$$2=2$$

(e) $y_{(t)}=8e^{7t}-1$

$$\frac{dy}{dt}=56e^{7t}$$

Substituting into equation

$$56e^{7t}-7(8e^{7t}-1)=7$$
$$56e^{7t}-56e^{7t}+7=7$$
$$7=7$$

(f) $y_{(t)}=\frac{5}{6}(1-e^{-2t})$

$$\frac{dy}{dt}=\frac{10}{6}e^{-2t}=\frac{5}{3}e^{-2t}$$

Substituting into the following differential equation

$$3\frac{dy}{dt}+6y=5$$
$$3\frac{5}{3}e^{-2t}+6\left[\frac{5}{6}-\frac{5}{6}e^{-2t}\right]=5$$
$$5e^{-2t}+5-5e^{-2t}=5$$
$$5=5$$

Dynamics of market price:

The framework:

Suppose that, for a particular commodity, the demand and supply functions are as follows

$$Q_d=\alpha-\beta P \qquad (\alpha,\beta>0)$$
$$Q_s=-\gamma+\delta P \qquad (\gamma,\delta>0)$$

At equilibrium $Q_d=Q_s$

$$\alpha-\beta P=-\gamma+\delta P$$
$$-\beta P-\delta P=-\alpha-\gamma$$
$$P(\beta+\delta)=\alpha+\gamma$$
$$\bar{P}=\frac{\alpha+\gamma}{\beta+\delta}$$

Let us assume, for the sake of simplicity, that the rate of price change (W.r.t time) at any moment is always directly proportional to the excess demand (Q_d-Q_s) prevailing at that moment. Such a pattern of change can be expressed symbolically as

$$\frac{dp}{dt}=j(Q_d-Q_s) \qquad (j>0)$$
$$\frac{dp}{dt}=j[(\alpha-\beta P)-(-\gamma+\delta P)]$$
$$\frac{dp}{dt}=j\alpha-j\beta P+j\gamma-j\delta P$$
$$\frac{dp}{dt}+j\beta P+j\delta P=j\alpha+j\gamma$$
$$\frac{dp}{dt}+j(\beta+\delta)P=j(\alpha+\gamma)$$

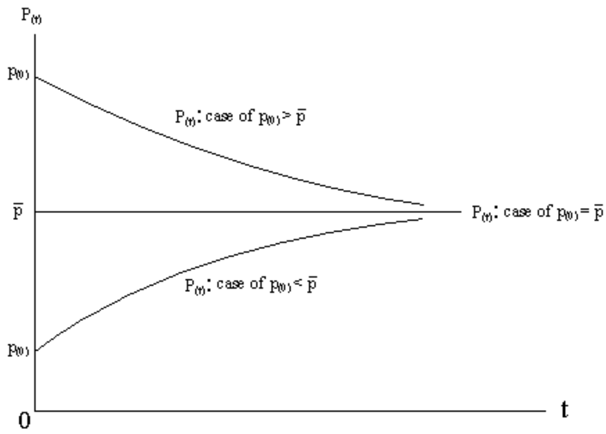
Since coefficient of P is non−zero we can apply the solution formula

$$P_{(t)}=\left[P_{(0)}-\frac{(\alpha+\gamma)}{(\beta+\delta)}\right]e^{-j(\beta+\delta)t}+\frac{\alpha+\gamma}{\beta+\delta}$$

$$P_{(t)}=[P_{(0)}-\bar{P}]e^{-j(\beta+\delta)t}+\bar{P}$$

$$P_{(t)}=[P_{(0)}-\bar{P}]e^{-kt}+\bar{P}$$

$$(Where \quad k=j(\beta+\delta))$$



The dynamic stability of equilibrium:

The first term on the right hand will tend to zero as $t\to\infty$. Since P $_{(0)}$ and $\bar{P}$ are both constant the key factor will be the exponential expression e^{-kt}. In view of the fact that k>0 that expression does tend to zero as $t\to\infty$ $P_{(t)}$ converges to the level of $\bar{P}$ which is the inter temporal equilibrium the equilibrium is said to be dynamically stable. Depending upon the relative magnitudes of P $_{(0)}$ and $\bar{P}$, the solution really encompasses three possible cases

(i) If $P_{(0)}=\bar{P}$, the first term on the right disappears and $P_{(t)}=\bar{P}$. The time path is a horizontal line and adjustment is immediate.

(ii) If $P_{(0)}>\bar{P}$, the first term on the right is positive. Thus $P_{(t)}>\bar{P}$ and $P_{(t)}$ approaches to $\bar{P}$ from above as $t\to\infty$ and first term on the right hand $\to 0$

(iii)If, $P_{(0)}<\bar{P}$ the first term on the right is negative and $P_{(t)}<\bar{P}$ and approaches it from below as $t\to\infty$ and the first term $\to 0$ as shown in figure.

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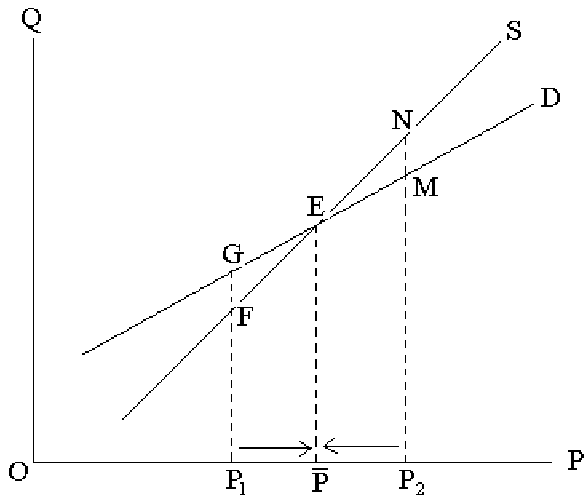
An alternative use of model:

In case the price adjustment is of the "normal" type, with j>0, so that excess demand drives price up rather than down then this restriction becomes merely $(\beta+\delta)>0$ or, equivalently,

$$\delta>-\beta$$

To have dynamic stability in that event, the slope of supply must exceed the slope of the demand. When both demand and supply are normally sloped $(-\beta<0,\ \delta>0)$, this requirement is obviously met. But even if one of the curve is sloped "perversely" the condition may still be full filled such as when $\delta=1$ and $-\beta=\frac{1}{2}$ (positively sloped demand)

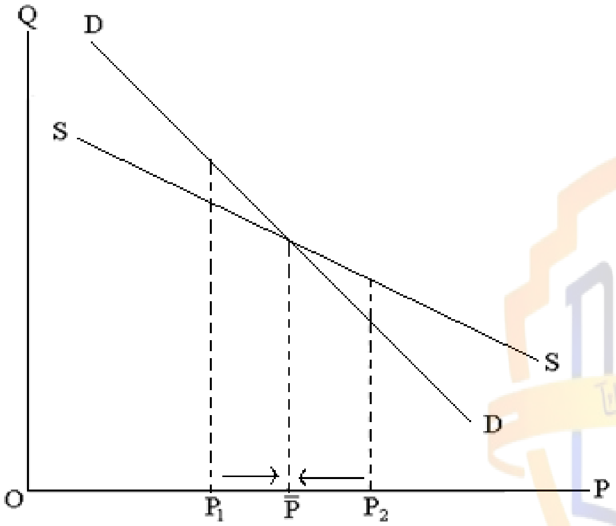


If the initial price happens to be at P_1 then Q_d (distance P_1G) will exceed Q_s

(distance P_1F) and excess demand will drive price up. On the other hand, if price is initially at P_2, then negative excess demand MN will drive price down. The price adjustment in either case is towards equilibrium.

Exercise 14.2:

Q. 1. If both the demand and supply in figure 14.2 is negatively slope instead, which curve should be steeper in order to have dynamic stability? Does your answer confirm to the criterion $\delta > -\beta$?



The demand curve should be steeper in order to have dynamic stability, i.e. $-\beta > \delta$. The answer does not confirm to the criterion $\delta > -\beta$

$Q_d = \alpha - \beta P$

$Q_\delta = -\gamma - \delta P$

$$\frac{dp}{dt} = j(Q_d - Q_s)$$

$$\frac{dp}{dt} = j(\alpha - \beta P + \gamma + \delta P)$$

$$\frac{dp}{dt} + j\beta P - j\delta P = j\alpha + j\gamma$$

$$\frac{dp}{dt} + j(\beta - \delta)P = j(\alpha + \gamma)$$

$$p_{(t)} = Ae^{-at} + \frac{b}{a}$$

$$p_{(t)} = Ae^{-j(\beta-\delta)} + \frac{j(\alpha+\gamma)}{j(\beta+\delta)}$$

$$p_{(t)} = Ae^{-j(\beta-\delta)t} + \frac{\alpha+\gamma}{\beta+\delta}$$

set $t = 0$

$$P_{(o)} = A + \frac{\alpha+\gamma}{\beta+\delta}$$

$$A = P_{(0)} - \frac{\alpha+\gamma}{\beta+\delta}$$

$$P_{(t)} = \left[P_{(0)} - \frac{\alpha+\gamma}{\beta+\delta}\right]e^{-j(\beta-\delta)t} + \frac{\alpha+\gamma}{\beta+\delta}$$

Since j>0, for dynamic stability $\beta - \delta > 0$

$\beta > \delta$

Hence the answer does not confirm to the criterion.

Q. 2. Show that (14.10) can be written as $\frac{dp}{dt} + k(p - \bar{p}) = 0$. If we let $(p - \bar{p}) \equiv \Delta$, so that $\frac{d\Delta}{dt} = \frac{dp}{dt}$, the differential equation can be further rewritten as

$$\frac{d\Delta}{dt} + k\Delta = 0$$

Find the time path and discuss the condition

For dynamic stability

Solution:

$$\frac{dp}{dt} + j(\beta+\delta)P = j(\alpha+\gamma)$$

Dividing through by $j(\beta+\delta)$

$$\frac{1}{j(\beta+\delta)}\frac{dp}{dt} + P = \frac{\alpha+\gamma}{\beta+\delta}$$

$$\frac{1}{k}\frac{dp}{dt} + P = \bar{P} \qquad [k = j(\beta+\delta)]$$

Rearranging the equation and multiplying through by k

$$\frac{dp}{dt} + k(p - \bar{p}) = 0$$

If we let $p - \bar{p} \equiv \Delta$ then $\frac{dp}{dt} = \frac{d\Delta}{dt}$

Substituting into above

$$\frac{d\Delta}{dt} + k\Delta = 0 \qquad (a = k, \quad b = 0)$$

$$\Delta_t = Ae^{-at} + \frac{b}{a}$$

$$\Delta_t = Ae^{-kt}$$

Since $k>0 \quad as \quad t\to\infty, \quad e^{-kt}\to 0 \quad \Delta t \to 0$

hence time path is dynamically stable.

Q.3: The dynamic market model discussed in this section is closely patterned after the static one in section 3.2. What specific new feature is responsible for transforming the static model into a dynamic one?

Answer: The assumption that the rate of price change at any moment is always directly proportional to excess demand $(Q_d - Q_s)$ symbolically

$\frac{dp}{dt} = j(Q_d - Q_s)$, is responsible for transforming the static model into dynamic model.

Q.4. let the demand and supply be

$$Q_d = \alpha - \beta P + \sigma \frac{dp}{dt} \qquad Q_s = -\gamma + \delta p$$

$$(\alpha,\ \beta,\ \gamma,\ \delta > 0)$$

(a) Assuming that the rate of change of price over time is directly proportional to the excess demand, find the time path $P_{(t)}$ (General solution)

$$\frac{dp}{dt} = j(Q_d - Q_s)$$

$$\frac{dp}{dt} = j\left[\left(\alpha - \beta P + \sigma\frac{dp}{dt}\right) - (-\gamma + \delta P)\right]$$

$$\frac{dp}{dt} = j\alpha - j\beta P + j\sigma\frac{dp}{dt} + j\gamma - j\delta P$$

$$\frac{dp}{dt} + j\beta P + j\delta P - j\sigma\frac{dp}{dt} = j\alpha + j\gamma$$

$$\frac{dp}{dt} - j\sigma\frac{dp}{dt} + j(\beta+\delta)P = j(\alpha+\gamma)$$

$$\frac{dp}{dt}(1-j\sigma) + j(\beta+\delta)P = j(\alpha+\gamma)$$

Dividing through by $(1-j\sigma)$

$$\frac{dp}{dt} + \frac{j(\beta+\delta)}{(1-j\sigma)}p = \frac{j(\alpha+\gamma)}{(1-j\sigma)}$$

Applying formula

$$p_{(t)} = Ae^{-at} + \frac{b}{a}$$

$$p_{(t)} = Ae^{\frac{-j(\beta+\delta)t}{1-j\sigma}} + \frac{j(\alpha+\gamma)/(1-j\sigma)}{j(\beta+\delta)/(1-j\sigma)}$$

$$p_{(t)} = Ae^{\frac{-j(\beta+\delta)t}{1-j\sigma}} + \frac{\alpha+\gamma}{\beta+\delta}$$

$$General \ \ Solution$$

(b) What is the inter temporal equilibrium price? What is market−clearing equilibrium price?

<u>Inter temporal equilibrium price:</u>

$\frac{\alpha+\gamma}{\beta+\delta}$ is inter temporal equilibrium price.

<u>Market−clearing equilibrium price:</u>

$$Q_d = Q_s$$

$$\alpha - \beta P + \sigma\frac{dp}{dt} = -\gamma + \delta P$$

$$-\beta P - \delta P = -\gamma - \alpha - \sigma\frac{dp}{dt}$$

$$\beta P + \delta P = \gamma + \alpha + \sigma\frac{dp}{dt}$$

$$p(\beta+\delta) = \gamma + \alpha + \sigma\frac{dp}{dt}$$

$$p = \frac{\alpha+\gamma}{\beta+\delta} + \frac{\sigma}{\beta+\delta}\frac{dp}{dt}$$

(c) What restriction on the parameter σ would ensure dynamic stability?

<u>Answer:</u>

Restriction on parameter σ that would ensure dynamic stability is

$\sigma \le 0$

Q.5. let the demand and supply be

$$Q_d = \alpha - \beta P - \eta\frac{dp}{dt} \qquad Q_s = \delta P$$

$$(\alpha,\ \beta,\ \eta,\ \delta > 0)$$

(a) Assuming that market is clear at every point of time, find the time path?

$$Q_d = Q_s$$

$$\alpha-\beta P-\eta\frac{dP}{dt}=\delta P$$

$$-\eta\frac{dp}{dt}-\beta P-\delta P=-\alpha$$

$$\eta\frac{dp}{dt}+\beta P+\delta P=\alpha$$

Dividing both sides by η

$$\frac{dp}{dt}+\frac{(\beta+\delta)}{\eta}p=\frac{\alpha}{\eta}$$

Applying formula to find time path of $P(P_t)$

$$P_{(t)}=Ae^{-at}+\frac{b}{a}$$

$$P_{(t)}=Ae^{\frac{-(\beta+\delta)t}{\eta}}+\frac{\alpha/\eta}{\beta+\delta/\eta}$$

$$P_{(t)}=Ae^{\frac{-(\beta+\delta)t}{\eta}}+\frac{\alpha}{\beta+\delta}$$

(b) Does this market have dynamically stable inter temporal equilibrium price?

Answer:

Yes this market have dynamically stable inter temporal equilibrium price because as $t\to\infty$

$$e^{-\frac{(\beta+\delta)t}{\eta}}\to 0 \quad Hence \;\; P_{(t)}\to\frac{\alpha}{\beta+\delta}$$

(c) The assumption of present model that $Q_d=Q_s$ for all t is identical with that of the static market model in sec.3.2. Nevertheless, we still have a dynamic model. How come?

Answer:

The model is dynamic due to the inclusion of time derivative $\frac{dp}{dt}$ in demand equation.

General formula for first order linear differential equations with variable coefficient and variable term:

In the more general case of a first order linear differential equation

$$\frac{dy}{dt}+u_{(t)}y=w_{(t)}$$

Where $u_{(t)}$ and $w_{(t)}$ represents variable coefficient and variable term, respectively. To find the time path $y_{(t)}$ in this case, the formula for a general solution is

$$y_{(t)}=e^{-\int udt}\left(A+\int we^{\int udt}dt\right)$$

Where A is an arbitrary constant, A solution is composed of two parts. $e^{-\int udt}$ A is called the complementary function and $e^{-\int udt}we^{\int udt}dt$ is called particular integral. The particular integral y_p equals the inter temporal equilibrium level of y_t; the complementary function y_c represents the deviation from equilibrium. For $y_{(t)}$ to be dynamically stable y_c must approach zero as t approaches infinity $(i.e.\ k\ \ in\ \ e^{kt}\ \ must\ \ be\ \ negative)$ the solution of a differential equation can always be checked by differential .

Note:

For the homogenous case, where $w_{(t)}=0$, the solution is still easy to obtain without using above formula, since the differential equation is in the form:

$$\frac{dy}{dt}+u_{(t)}y=0 \quad or \quad \frac{1}{y}\frac{dy}{dt}=-u_{(t)}$$

By integrating both sides W.r.t “t”

$$Left\ \ side=\int\frac{1}{y}\frac{dy}{dt}dt=\int\frac{dy}{y}=\ln y+c$$

(Assuming y > 0)

$$Right\ \ side=\int -u_{(t)}dt=-\int u_{(t)}dt$$

When the two sides are equated, the result is

$$\ln y=-c-\int u_{(t)}dt$$

By taking antilog, we can obtain $y_{(t)}$

$$e^{l\,ny}=e^{-c}e^{-\int u(t)dt}$$

$$y=Ae^{-\int u(t)dt} \qquad (Where\ \ A\equiv e^{-c})$$

Example 1:

Find the general solution of the equation

$$\frac{dy}{dt}+3t^2y=0$$

Solution:

$$As\ \ w_{(t)}=0$$

$$u_{(t)}=3t^2$$

Applying formula $y_{(t)}=Ae^{-\int u(t)dt}$ to find general solution

$$\int udt=\int 3t^2dt=t^3+c$$

Hence $y_{(t)}$ will be

$$y_{(t)}=Ae^{-\int udt}=Ae^{-t^3}e^{-c}=Be^{-t^3}$$

$$(Where\ \ B=Ae^{-c})$$

Example 2:

Find the general solution of the equation

$$\frac{dy}{dt}+2ty=t$$

Solution:

Here we have

$$u=2t \quad w=t \quad and \quad \int udt=t^2+k$$

$$(k\ \ arbitrary)$$

Applying formula

$$y_{(t)}=e^{-\int udt}(A+\int we^{\int udt}dt)$$

$$y_{(t)}=e^{-(t^2+k)}(A+\int te^{t^2+k}dt)$$

$$y_{(t)}=e^{-t^2}e^{-k}(A+e^k\int te^{t^2}dt)$$

$$Let\ \ u=t^2$$

$$\frac{du}{dt}=2t\Rightarrow \quad dt=\frac{du}{2t}$$

Substituting into above

$$y_{(t)}=e^{-t^2}e^{-k}(A+e^k\int te^u.\ \frac{du}{2t})$$

$$y_{(t)}=e^{-t^2}e^{-k}(A+\frac{1}{2}e^k\int e^u.du)$$

$$y_{(t)}=e^{-t^2}e^{-k}(A+\frac{1}{2}e^ke^u+c)$$

$$y_{(t)}=e^{-t^2}e^{-k}A+\frac{1}{2}e^ke^u.\ e^{-k}(e^{t^2}+c)$$

$$As\ \ u=t^2$$

$$y_{(t)}=Ae^{-k}.\ e^{-t^2}+e^{-t^2}\frac{1}{2}e^{t^2}+c^{-t^2})$$

$$y_{(t)}=(Ae^{-k}+c)e^{-t^2}+\frac{1}{2}$$

$$Where\ \ B=Ae^{-k}+c)$$

$$y_{(t)}=Be^{-t^2}+\frac{1}{2}$$

Example 3:

$$\frac{dy}{dt}+4ty=4t$$

We have

$$u=4t$$

$$w=4t$$

$$y_{(t)}=e^{-\int udt}(A+\int we^{\int udt}dt)$$

$$y_{(t)}=e^{-\int 4tdt}(A+\int 4te^{\int 4tdt}dt)$$

$$y_{(t)}=e^{-2t^2}(A+\int 4te^{2t^2}dt)$$

$$Let\ \ u=2t^2$$

$$\frac{du}{dt}=4t \ \Rightarrow\ dt=\frac{du}{4t}$$

$$y_{(t)}=e^{-2t^2}(A+\int 4te^u.\ \frac{du}{4t})$$

$$y_{(t)}=e^{-2t^2}(A+\int e^u.\ du)$$

$$y_{(t)}=e^{-2t^2}(A+e^u+c)=e^{-2t^2}(A+e^{2t^2}+c)$$

$$y_{(t)}=Ae^{-2t^2}+e^{2t^2}e^{-2t^2}+ce^{-2t^2}$$

$$y_{(t)}=Ae^{-2t^2}+e^0+ce^{-2t^2}$$

$$y_{(t)}=(A+c)e^{-2t^2}+1 \qquad As\ \ e^0=1)$$

$$y_{(t)}=Be^{-2t^2}+1 \qquad (B=A+c)$$

Exercise 14.3:

Q.1. Solve the following first order linear differential equation, if an initial condition is given definitize the arbitrary constant.

1. $\frac{dy}{dt}+5y=15$

$u=5, \quad w=15 \quad and \quad \int udt=\int 5dt=5t+c$

Applying formula

$y_{(t)}=e^{-\int udt}(A+\int we^{\int udt}dt)$

$y_{(t)}=e^{-5t}.\ e^{-c}(A+\int 15e^{5t+c}dt)$

$y_{(t)}=e^{-5t}.\ e^{-c}(A+\int 15e^{5t}e^{c}dt)$

$y_{(t)}=e^{-5t}.\ e^{-c}(A+15e^{c}\int e^{5t}dt)$

$y_{(t)}=e^{-5t}.\ e^{-c}\left[A+15e^{c}(\frac{e^{5t}}{5}+c)\right]$

$y_{(t)}=e^{-5t}.\ e^{-c}[A+3e^{5t}e^{c}+15e^{c}c]$

$y_{(t)}=e^{-5t}.\ e^{-c}A+3e^{5t}e^{c}e^{-5t}e^{-c}+15e^{c}ce^{-5t}e^{-c}$

$y_{(t)}=e^{-5t}(Ae^{-c}+15c)+3$

$y_{(t)}=Be^{-5t}+3 \qquad \because\ (Ae^{-c}+15c=B)$

2. $\frac{dy}{dt}+2ty=0$

$Since \ w=0 \ and \ u=2t$, we will try the following formula to get solution

$y_{(t)}=Ae^{\int udt}$

$y_{(t)}=Ae^{-\int 2tdt}=Ae^{-(t^2+c)}=Ae^{-t^2}e^{-c}$

$y_{(t)}=Ae^{-c}.\ e^{-t^2}=Be^{-t^2}$

$(Ae^{-c}=B)$

3. $\frac{dy}{dt}+2ty=t \qquad y_{(0)}=\frac{3}{2}$

$u=2t$

$w=t \quad and \quad \int udt=\int 2tdt=t^2+c$

$y_{(t)}=e^{-(t^2+c)}(A+\int te^{t^2+c}dt)$

$y_{(t)}=e^{-(t^2+c)}(A+e^{c}\int te^{t^2}dt)$

$y_{(t)}=e^{-t^2}.\ e^{-c}\left\{A+e^{c}(\frac{1}{2}e^{t^2}+c)\right\}$

$y_{(t)}=e^{-t^2}.\ e^{-c}A+e^{-t^2}.\ e^{-c}.\ e^{c}(\frac{1}{2}e^{t^2}+c)$

$y_{(t)}=e^{-t^2}.\ e^{-c}A+\frac{1}{2}e^{t^2}e^{-t^2}+c\ e^{-t^2}$

$y_{(t)}=Ae^{-c}.\ e^{-t^2}+\frac{1}{2}+ce^{-t^2}$

$y_{(t)}=e^{-t2}(Ae^{-c}+c)+\frac{1}{2}$

$y_{(t)}=Be^{-t2}+\frac{1}{2} \qquad (B=Ae^{-c}+c)$

Note:

Recall rule II a in chapter B for integrating $\int te^{t^2}$

To definitize the solution set t=0

$y_{(0)}=Be^{0}+\frac{1}{2}$

$\frac{3}{2}=B+\frac{1}{2}$

$B=\frac{3}{2}-\frac{1}{2}=\frac{2}{2}=1$

Hence

$y_{(t)}=e^{-t^2}+\frac{1}{2} \qquad Definite\ \ solution$

Q. 4. $\frac{dy}{dt}+t^2y=5t^2 \qquad y_{(0)}=6$

$u=t^2 \quad w=5t^2 \quad and \quad \int udt=\int t^2dt=\frac{1}{3}t^3+c$

Applying formula for general solution

$y_{(t)}=e^{-\int udt}(A+\int we^{\int udt}dt)$

$y_{(t)}=e^{-(\frac{1}{3}t^3+c)}(A+\int 5t^2e^{\frac{1}{3}t^3+c}dt)$

$y_{(t)}=e^{-\frac{1}{3}t^3}e^{-c}(A+\int 5t^2e^{\frac{1}{3}t^3}e^{c}dt)$

$y_{(t)}=e^{-\frac{1}{3}t^3}e^{-c}(A+5e^{c}\int t^2e^{\frac{1}{3}t^3}dt)$

$y_{(t)}=e^{-\frac{1}{3}t^3}e^{-c}\left[A+5e^{c}(e^{\frac{1}{3}t^3}+c)\right]$

$By\ \ rule\ \ IIa$

$y_{(t)}=e^{-\frac{1}{3}t^3}e^{-c}A+5e^{-\frac{1}{3}t^3}(e^{\frac{1}{3}t^3}+c)$

$y_{(t)}=Ae^{-c}e^{-\frac{1}{3}t^3}5+5e^{-\frac{1}{3}t^3}.\ c$

$y_{(t)}=(Ae^{-c}+5c)e^{-\frac{1}{3}t^3}+5$

$y_{(t)}=Be^{-\frac{1}{3}t^3}+5 \qquad (B=Ae^{-c}+5c)$

$Set\ \ t=0$, to definitize the solution

$y_{(0)}=Be^{0}+5$

$y_{(0)}=B+5 \qquad (e^0=1)$

$6=B+5 \Rightarrow B=6-5=1$

Substituting into above general solution

$y_{(t)}=e^{\frac{-1}{3}t^3}+5$

5. $2\frac{dy}{dt}+12y+2e^t=0 \qquad y_{(0)}=\frac{6}{7}$

Dividing through by 2 and rearranging the equation

$\frac{dy}{dt}+6y=-e^t$

$u=6 \quad w=-e^t \quad and \quad \int udt=\int 6dt=6t+c$

Applying formula for general solution

$y_{(t)}=e^{-\int udt}(A+\int we^{\int udt}dt)$

$y_{(t)}=e^{-(6t+c)}(A+\int -e^t e^{(6t+c)}dt)$

$y_{(t)}=e^{-(6t+c)}(A+\int -e^t e^{6t}.\ e^c dt)$

$y_{(t)}=e^{-(6t+c)}(A-e^c\int e^t e^{6t}dt)$

$y_{(t)}=e^{-(6t+c)}(A-e^c\int e^{7t}dt)$

$y_{(t)}=e^{-6t}Ae^{-c}-e^{-6t}(\frac{1}{7}e^{7t}+c)$

$y_{(t)}=e^{-6t}Ae^{-c}-\frac{1}{7}e^t-ce^{-6t}$

$y_{(t)}=(Ae^{-c}-c)e^{-6t}-\frac{1}{7}e^t$

$y_{(t)}=Be^{-6t}-\frac{1}{7}e^t \qquad (B=Ae^{-c}-c)$

$As \quad y_{(0)}=\frac{6}{7} \quad set \quad t=0$

$y_{(0)}=B-\frac{1}{7}e^0$

$\frac{6}{7}=B-\frac{1}{7} \qquad (e^0=1)$

$\frac{6}{7}+\frac{1}{7}=B \quad \Rightarrow \quad B=\frac{7}{7}=1$

Hence

$y_{(t)}=e^{-6t}-\frac{1}{7}e^t$

6. $\frac{dy}{dt}+y=t$

$u=1 \quad w=t \quad and \quad \int udt=\int dt=t+c$

$y_{(t)}=e^{-\int udt}(A+\int we^{\int udt}dt)$

$y_{(t)}=e^{-(t+c)}(A+\int te^{t+c}dt)$

For

$\int te^t dt=?$

$Let \quad f(t)=t \quad g'(t)=e^t$

$f'(t)=1 \quad g(t)=\int e^t dt=e^t$

$\int te^t dt=te^t-\int e^t dt$

$\int te^t dt=te^t-e^t=e^t(t-1)+c$

$y_{(t)}=e^{-t}e^{-c}(A+e^c[e^t t-e^t+c])$

$y_{(t)}=e^{-t}Ae^{-c}+t-1+e^{-t}c$

$y_{(t)}=e^{-t}(Ae^{-c}+c)+(t-1)$

$y_{(t)}=Be^{-t}+t-1 \qquad (B=Ae^{-c}+c)$

$General \quad solution$

4: Exact Differential Equation:

Given a function of more than one independent variable, such as $F(y,t)$ where $M=\partial F/\partial y$ and $N=\partial F/\partial t$, the total differential is written

$dF(y,t)=Mdy+Ndt$

Since F is a function of more than one independent variable, M and N are partial derivation and equation is called a partial differential equation. If the differential is set equal to zero, so that $Mdy+Ndt=0$, it is called an exact differential equation because the left side exactly equals the differential of the primitive function $F(y,t)$. For an exact differential equation, $\partial M/\partial t$ must equal $\partial N/\partial y$ that is,

$\frac{\partial^2 F}{\partial t\partial y}=\frac{\partial^2 F}{\partial y\partial t} \qquad (By \ Young's \ theorem)$

Solution of an exact differential equation calls for successive integration with respect to one independent variable at a time while holding constant the other independent variable(s)

$F(y,t)=\int M dy+\psi(t)$

Or

$F(y,t)=\int N dt+\psi(y)$

Example 1:

Solve the exact differential equation

$2ytdy+y^2dt=0$

$M=2yt$

$N=y^2$

$(i)\quad \frac{\partial M}{\partial t}=2y=\frac{\partial N}{\partial y}$

$(ii)\ F(y,t)=\int 2yt\,dy+z(t)$

$F(y,t)=y^2t+z(t)$

$\frac{\partial F}{\partial t}=y^2+z'(t) \qquad \left(\frac{\partial F}{\partial t}=N=y^2\right)$

$y^2=y^2+z'(t)$

$z'(t)=y^2-y^2=0$

$(iv)\ z(t)=\int 0\,dt=k \qquad k$ constant

$(v)\ F(y,t)=y^2t+k$

Example 2:

Solve the equation

$(t+2y)dy+(y+3t^2)dt=0$

Solution:

$M=t+2y \qquad N=y+3t^2$

$(i)\ \frac{\partial M}{\partial t}=1=\frac{\partial N}{\partial y}$

$(ii)\ F(y,t)=\int M dy+\psi(t)$

$F(y,t)=\int (t+2y)dy+\psi(t)$

$F(y,t)=ty+y^2+\psi(t)$

$(iii)\ \frac{\partial F}{\partial t}=y+\psi'(t)$

$y+3t^2=y+\psi'(t) \qquad (\frac{\partial F}{\partial t}=N=y+3t^2)$

$\psi'(t)=3t^2$

$(iv)\ \psi(t)=\int 3t^2dt=t^3+c$

$(v)\ F(y,t)=ty+y^2+t^3+c$

Example 3:

The differential equation

$2tdy+ydt=0$

Is not exact because:

$\frac{\partial M}{\partial t}=\frac{\partial(2t)}{\partial t}=2\neq\frac{\partial N}{\partial y}=\frac{\partial(y)}{\partial y}=1$

Cross partial derivatives are not equal. When the cross partial derivatives are not equal, the equation is not exact. However some equations can be made exact by means of an integrating factor.

Integrating factor:

This is a multiple which permits the equation to be integrated.

Rules for the integrating factor:

Two rules will help to find the integrating factor for a nonlinear first order differential equation if such a factor exists. Assuming $\partial M/\partial t\neq\partial N/\partial y$

Rule 1:

If $\frac{1}{N}\left(\frac{\partial M}{\partial t}-\frac{\partial N}{\partial y}\right)=f(y)$ alone, then $e^{\int f(y)dy}$ is an integrating factor.

Rule 2:

If $\frac{1}{M}\left(\frac{\partial N}{\partial y}-\frac{\partial M}{\partial t}\right)=g(t)$ alone, then $e^{\int g(t)dt}$ is an integrating factor.

Example (a):

Find the integrating factor for the differential equation given below (b) solve the equation, using the four step procedure.

$(7y+4t^2)dy+4tydt=0$

$M=7y+4t^2 \qquad N=4ty$

$\frac{\partial M}{\partial t}=8t\neq\frac{\partial N}{\partial y}=4t$

Applying rule 1 to find the integrating factor

$\frac{1}{N}(\frac{\partial M}{\partial t}-\frac{\partial N}{\partial y})=\frac{1}{4ty}(8t-4t)=\frac{4t}{4ty}=\frac{1}{y}$

$=f(y)\ alone$

Thus the integrating factor is

$e^{\int \frac{1}{y}dy}=e^{\ln y}=y$

Multiplying given equation by y

$(7y^2+4yt^2)dy+4ty^2dt=0$

$\frac{\partial M}{\partial t}=8yt=\frac{\partial N}{\partial y}$ thus

1. $F(y,t)=\int(7y^2+4yt^2)dy+\psi(t)$

$F(y,t)=\frac{7}{3}y^3+2y^2t^2+\psi(t)$

2. $\frac{\partial F}{\partial t}=4y^2t+\psi'(t)$

$4y^2t=4y^2t+\psi'(t)$ $\quad (\frac{\partial F}{\partial t}=N=4y^2t)$

$\psi'(t)=0$

3. $\psi(t)=\int\psi'(t)dt=\int 0=k$

4. $F(y,t)=\frac{7}{3}y^3+2y^2t^2+k$ $\quad k$ constant

$F(y,t)=7y^3+6y^2t^2+k$

Assignment:

Solve yourself

$(1)\ y^3tdy+\frac{1}{2}y^4dt=0$

$(2)\ 4tdy+(16y-t^2)dt=0$

$(3)\ t^2dy+3ytdt=0$

$(4)\ (y-t)dy-dt=0$

Solution of first order linear differential equations:

The general first−order linear differential equation

$\frac{dy}{dt}+uy=w$

$\frac{dy}{dt}+uy-w=0$

$dy+(uy-w)dt=0$

$M=1 \qquad N=uy-w$

$\frac{\partial M}{\partial t}=0 \quad \neq \quad \frac{\partial N}{\partial y}=u$

Applying formula to find the integrating factor

$\frac{1}{M}(\frac{\partial N}{\partial y}-\frac{\partial M}{\partial t})=f(t)\ alone \rightarrow e^{\int f(t)dt}$

$\frac{1}{1}(u-0)=u\ alone \rightarrow e^{\int udt}$

Hence integrating factor is $e^{\int udt}$ multiplying equation by $e^{\int udt}$

$e^{\int udt}dy+e^{\int udt}(uy-w)dt=0$

$F(y,t)=c$

(i). $F(y,t)=\int Mdy+\psi(t)$

$F(y,t)=\int e^{\int udt}dy+\psi(t)$

$F(y,t)=ye^{\int udt}+\psi(t)$

(ii). $\frac{\partial F}{\partial t}=yue^{\int udt}+\psi'(t)$

$N=yue^{\int udt}+\psi'(t)$

$e^{\int udt}(uy-w)=yue^{\int udt}+\psi'(t)$

$\psi'(t)=e^{\int udt}(uy-w)-yue^{\int udt}$

$\psi'(t)=e^{\int udt}uy-we^{\int udt}-yue^{\int udt}$

$\psi'(t)=-we^{\int udt}$

(iii) $\psi(t)=-\int we^{\int udt}dt$

(iv) $F(y,t)=ye^{\int udt}-\int we^{\int udt}dt$

Since $F(y,t)=c$

Therefore $ye^{\int udt}-\int we^{\int udt}dt=c$

$ye^{\int udt}=c+\int we^{\int udt}dt$

$y=\frac{c+\int we^{\int udt}dt}{e^{\int udt}}$

$y_{(t)}=e^{-\int udt}(A+\int we^{\int udt}dt) \quad (\because A=c)$

Exercise 14.4:

1. Verify that each of the differential equation is exact and solve by the four step procedure.

(a) $2yt^3dy+3y^2t^2dt=0$

$M=2yt^3 \qquad N=3y^2t^2$

$\frac{\partial M}{\partial t}=6yt^2=\frac{\partial N}{\partial y}$ *The equation is exat. Applying four step procedure*

Step I:

$F(y,t)=\int Mdy+\psi(t)$

$F(y,t)=\int 2yt^3dy+\psi(t)$

$F(y,t)=y^2t^3+\psi(t)$

Step II:

$\frac{\partial F}{\partial t}=3y^2t^2+\psi'(t)$

$3y^2t^2=3y^2t^2+\psi'(t) \qquad \left(\frac{\partial F}{\partial t}=N=3y^2t^2\right)$

$\psi'(t)=0$

Step III:

$\psi(t)=\int \psi'(t)dt=k$

Step IV:

$F(y,t)=y^2t^3+k$

$y^2=k$

$y^2=\frac{k}{t^3}$

$y=\left(\frac{c}{t^3}\right)^{\frac{1}{2}} \qquad (k=c)$

(b) $3y^2tdy+(y^3+2t)dt=0$

$M=3y^2t \qquad N=y^3+2t$

$\frac{\partial M}{\partial t}=3y^2= \qquad \frac{\partial N}{\partial y}=3y^2$

The equation is exact. Applying four−step procedure

(i) $F(y,t)=\int 3y^2tdy+\psi(t)$

$F(y,t)=y^3t+\psi(t)$

(ii) $\frac{\partial F}{\partial t}=y^3+\psi'(t)$

$y^3+2t=y^3+\psi'(t)$

$\psi'(t)=2t$

(iii) $\psi(t)=\int \psi'(t)dt=\int 2tdt=t^2+c$

(iv) $F(y,t)=y^3t+t^2+c$

$y^3=\frac{-t^2-c}{t}$

$y=\left(\frac{-t^2-c}{t}\right)^{\frac{1}{3}}$

(c) $t(1+2y)dy+y(1+y)dt=0$

$M=t+2yt \qquad N=y+y^2$

$\frac{\partial M}{\partial t}=1+2y=\frac{\partial N}{\partial y}=1+2y$

The equation is exact. Applying four−step procedure

(i) $F(y,t)=\int (t+2yt)dy+\psi(t)$

$F(y,t)=ty+y^2t+\psi(t)$

(ii) $\frac{\partial F}{\partial t}=y+y^2+\psi'(t)$

$y+y^2=y+y^2+\psi'(t)$

$\left(\frac{\partial F}{\partial t}=N=y+y^2\right)$

$\psi'(t)=0$

(iii) $\psi(t)=\int \psi'(t)dt=c$

(iv) $F(y,t)=ty+y^2t+c$

$yt+y^2t=c$

(d) $\frac{dy}{dt}+\frac{2y^4t+3t^2}{4y^3t^2}=0$

$(4y^3t^2)dy+(2y^4t+3t^2)dt=0$

$M=4y^3t^2 \qquad N=2y^4t+3t^2$

$\frac{\partial M}{\partial t}=8y^3t \qquad \frac{\partial N}{\partial y}=8y^3t$

The equation is exact. Applying four−step procedure

(i) $F(y,t)=\int 4y^3t^2dy+\psi(t)$

$F(y,t)=y^4t^2+\psi(t$

(ii) $\frac{\partial F}{\partial t}=2y^4t+\psi'(t)$

$2y^4t+3t^2=2y^4t+\psi'(t)$

$\psi'(t)=3t^2$

(iii) $\psi(t)=\int \psi'(t)dt=\int 3t^2dt=t^3+c$

(iv) $F(y,t)=y^4t^2+t^3+c$

2. Are the following differential equations are exact? If not, try t, y and y^2 as possible integrating factor.

(a) $2(t^3+1)dy+3yt^2dt=0$

$2t^3+2=M \qquad N=3yt^2$

$\frac{\partial M}{\partial t}=6t^2 \quad \neq \quad \frac{\partial N}{\partial y}=3t^2$

Trying 't' 'y' and 'y^2' respectively

(i) $M=2t^4+2t \qquad N=3yt^3$

$\frac{\partial M}{\partial t}=8t^3+2\neq\frac{\partial N}{\partial y}=3t^3$

(ii) $M=2t^3y+2y \qquad N=3y^2t^2$

$\frac{\partial M}{\partial t}=6yt^2=\frac{\partial N}{\partial y}=6yt^2$

(iii) $M=2t^3y^2 \qquad N=3y^3t^2$

$\frac{\partial M}{\partial t}=6t^2y^2 \quad \neq \quad \frac{\partial N}{\partial y}=9y^2t^2$

Hence possible integrating factor is "y"

(b) $4y^3tdy+(2y^4+3t)dt=0$

$M=4y^3t \qquad N=2y^4$

$\frac{\partial M}{\partial t}=4y^3 \quad \neq \quad \frac{\partial N}{\partial y}=8y^3$

Trying 't' 'y' and 'y^2' as to find possible integrating factor trying t

$4y^3t^2dy+(2y^4t+3t^2)dt=0$

$M=4y^3t^2 \qquad N=2y^4t+3t^2$

$\frac{\partial M}{\partial t}=8y^3t \quad \neq \quad \frac{\partial N}{\partial y}=8y^3t$

Trying y

$4y^4tdy+(2y^5+3ty)dt=0$

$M=4y^4t \qquad N=2y^5+3ty$

$\frac{\partial M}{\partial t}=4y^4 \quad \neq \quad \frac{\partial N}{\partial y}=10y^4+3t$

Trying y^2

$4y^5tdy+(2y^6+3ty^2)dt=0$

$M=4y^5t \qquad N=2y^6+3ty^2$

$\frac{\partial M}{\partial t}=4y^5 \quad \neq \quad \frac{\partial N}{\partial y}=12y^5+6ty$

Hence possible integrating factor is "t"

3. By applying the four step procedure to the general exact differential equation $Mdy+Ndt=0$ derive the following formula for the general solution of an exact differential equation

$$\int Mdy+\int Ndt-\int\left(\frac{\partial}{\partial t}\int Mdy\right)dt=c$$

As the general form of exact differential equation is

$Mdy+Ndt=0$

$F(y,t)=c$

(i) $F(y,t)=\int Mdy+\psi(t)$

(ii) $\frac{\partial F}{\partial t}=\frac{\partial}{\partial t}\int Mdy+\psi'(t)$

$\psi'(t)=N-\frac{\partial}{\partial t}\int Mdy \qquad \left(\frac{\partial F}{\partial t}=N\right)$

(iii) $\psi(t)=\int Ndt-\int\left(\frac{\partial}{\partial t}\int Mdy\right)dt$

(iv) $F(y,t)=\int Mdy+\int Ndt-\int\left(\frac{\partial}{\partial t}\int Mdy\right)d(t)$

Since $F(y,t)=c$

Therefore

$$\int Mdy+\int Ndt-\int\left(\frac{\partial}{\partial t}\int Mdy\right)d(t)=c$$

Nonlinear Differential Equation of the First Degree:

Solution of nonlinear first−order, first degree differential equation is complex (A first order first degree differential equation is one in which the highest derivative is the first derivative $\frac{dy}{dt}$ and that derivative is raised to a power of 1. It is nonlinear if it contains a product of y and $\frac{dy}{dt}$ or y raised to power other than one) In general an equation in the form

$f(y,t)dy+g(y,t)dt=0$

Or

$\frac{dy}{dt}=h(y,t)$

Where there is no restriction on the power of y and t, constitutes a first−order first−degree nonlinear differential equation. In this context, we shall discuss three cases

Case 1: Exact differential equation:

This case is discussed earlier.

Case 2: Separable variables:

If the equation is exact or can be rendered exact by an integrating factor, the procedure out lined above, i.e. four step procedures can be used. If however the equation can be written in the form of separated variables such that

$$f(y)dy+g(t)dt=0$$

Where f and g respectively are the functions of y and t alone, the equation can be solved simply by ordinary integration. This procedure is illustrated in the following examples

Example 1:

Solve the equation

$$3y^2dy-tdt=0$$

First let us rewrite the equation

$$3y^2dy=tdt$$

Integrating both sides

$$\int 3y^2dy=\int tdt \quad or \quad y^3+c_1=\frac{1}{2}t^2+c_2$$

$$y^3=\frac{1}{2}t^2+c \quad or \quad y(t)=\left(\frac{1}{2}t^2+c\right)^{\frac{1}{3}}$$

Example 2:

Solve the equation

$$\frac{1}{y}dy+\frac{1}{2t}dt=0$$

Integrating both sides

$$\int\frac{1}{y}dy+\int\frac{1}{2t}dt=\int 0$$

$$\ln y+\frac{1}{2}\ln t=c \quad or \quad \ln\left(yt^{\frac{1}{2}}\right)=c$$

Taking antilog

$$e^{\ln\left(yt^{\frac{1}{2}}\right)}=e^c$$

$$yt^{\frac{1}{2}}=e^c \;\Rightarrow\; yt^{\frac{1}{2}}=k \;\Rightarrow\; y=kt^{\frac{-1}{2}}$$

Case III Bernoulli equation:

If the differential equation $\frac{dy}{dt}=h(y,t)$ happens to take the specific nonlinear form

$$\frac{dy}{dt}+Ry=Ty^m$$

Where R and T two functions of t, and m is any number other than zero and 1(what if m=0 or m=1?) then the equation−referred to as a Bernoulli equation can always be reduced to a linear differential equation and solved as such. First divide the above equation by y^m to get

$$y^{-m}\frac{dy}{dt}+Ry^{1-m}=T$$

$$Let\ z=y^{1-m} \quad then \quad \frac{dz}{dt}=\frac{dz}{dy}\frac{dy}{dt}$$

$$=(1-m)y^{-m}\frac{dy}{dt}$$

Then the preceding equation can be written as

$$\frac{1}{1-m}\frac{dz}{dt}+Rz=T$$

Multiplying through by (1−m)dt and rearranging, we can than form the equation into

$$dz+[(1-m)Rz-(1-m)T]dt=0$$

This is first order linear differential equation in variable

2. We can apply formula to find it, solution (zt)

$$z(t)=e^{-\int udt}\left(A+\int we^{\int udt}dt\right)$$

Then as a final step we can translate z back to y by reverse substitution.

Example 3:

Solve the equation

$$\frac{dy}{dt}+ty=3ty^2$$

This is Bernoulli equation in variable y with m=2

As $m=2 \quad let \quad y^{1-m}=z \;\Rightarrow\; z=y^{1-2}=y^{-1}$

$R=t \qquad T=3t$

Substituting into the formula

$$\frac{dz}{dt}+(1-m)Rz=(1-m)T$$

$$\frac{dz}{dt}+(1-2)tz=(1-2)3t$$

$$\frac{dz}{dt}-tz=-3t$$

Applying formula to solve the linear differential equation in variable z

$$z_{(t)}=e^{-\int udt}\left(A+\int we^{\int udt}dt\right)$$

$$z_{(t)}=e^{-\int -tdt}\left(A+\int -3te^{\int -tdt}dt\right)$$

$$z_{(t)}=e^{\frac{1}{2}t^2}\left(A-\int 3te^{\frac{-1}{2}t^2}dt\right)$$

$Let \quad u=\frac{-1}{2}t^2$

$$\frac{dz}{dt}=-t \;\Rightarrow\; dt=\frac{du}{-t}$$

$$z_{(t)}=e^{\frac{1}{2}t^2}\left(A-3\int te^u\frac{du}{-t}\right)$$

$$z_{(t)}=e^{\frac{1}{2}t^2}\left(A+3\int e^u\, du\right)$$

$z_{(t)}=e^{\frac{1}{2}t^2}\left(A+3e^u\right)$ Ignoring constant of integration

As $u=\frac{-1}{2}t^2$

$$z_{(t)}=e^{\frac{1}{2}t^2}\left(A+3e^{\frac{-1}{2}t^2}\right)$$

$$z_{(t)}=Ae^{\frac{1}{2}t^2}+3$$

As $z(t)=y_{(t)}^{-1}$

$$y_{(t)}^{-1}=Ae^{\frac{1}{2}t^2}+3$$

$$y_{(t)}=\left(Ae^{\frac{1}{2}t^2}+3\right)^{-1}$$

Example 4:

Solve the equation

$$\frac{dy}{dt}+\frac{1}{t}y=y^3$$

$$\frac{dy}{dt}+\frac{1}{t}y=y^3$$

$m=3 \quad let \quad z=y^{1-m}=y^{1-3}=y^{-2}$

$R=\frac{1}{t} \;,\; T=1$

Substituting into formula

$$\frac{dz}{dt}+(1-m)Rz=(1-m)T$$

$$\frac{dz}{dt}+(1-3)\frac{1}{t}z=(1-3)1$$

$$\frac{dz}{dt}+\frac{-2}{t}z=-2$$

$$\frac{dz}{dt}-\frac{2}{t}z=-2$$

This first order differential equation in variable z where $u=\frac{-2}{t} \quad w=-2$

Applying formula for the general solution

$$z_{(t)}=e^{-\int udt}\left(A+\int we^{\int udt}dt\right)$$

$$z_{(t)}=e^{-\int\frac{2}{t}dt}\left(A+\int -2e^{-\int\frac{2}{t}dt}dt\right)$$

$$z_{(t)}=e^{2\ln t}\left(A-2\int e^{-2\ln t}dt\right)$$

$$z_{(t)}=e^{\ln t^2}\left(A-2\int e^{\ln t^{-2}}dt\right)$$

$$z_{(t)}=t^2\left(A-2\int t^{-2}dt\right)$$

$$z_{(t)}=t^2\left(A+2t^{-1}\right)= At^2+2t$$

$As \quad z=y^{-2}$

$$y^{-2}=At^2+2t$$

$$y_{(t)}=\left(At^2+2t\right)^{\frac{-1}{2}}$$

Exercise 14.5:

1. Determine for each of the following (i) whether the variables are separable and (ii) whether the equation is linear or else can be linearized:

(a) $2tdy+2ydt=0$

(i) Dividing both sides by 2ty

$$\frac{2tdy}{2ty}+\frac{2ydt}{2ty}=0$$

$\frac{1}{y}dy+\frac{1}{t}dt=0$

Yes the variables are separable

(ii) $2tdy+2ydt=0$

$2tdy=-2ydt$

$\frac{dy}{dt}=\frac{-2y}{2t}$

$\frac{dy}{dt}+\frac{1}{t}y=0$

Yes the equation is linear

(b) $\frac{y}{y+t}dy+\frac{2t}{y+t}dt=0$

(i) Multiplying both sides by $y+t$

$ydy+2tdt=0$

Hence the variables are separable

(ii) $\frac{y}{y+t}dy+\frac{2t}{y+t}dt=0$

Multiplying both sides by y + t

$ydy+2tdt=0$

$ydy=-2tdt$

$\frac{dy}{dt}=-\frac{2t}{y}$

$\frac{dy}{dt}=-2ty^{-1}$

The equation is non linear (i.e. Bernoulli equation) and can be linearized.

(c) $\frac{dy}{dt}=\frac{-t}{y}$

(i) By rearranging

$ydy=-tdt$

$ydy+tdt=0$

Yes variables are separable

(ii) $\frac{dy}{dt}=-\frac{t}{y}$

$\frac{dy}{dt}=-ty^{-1}$

Equation is not linear but can be linearized as Bernoulli equation

(d) $\frac{dy}{dt}=3y^2t$

(i) $\frac{1}{y^2}dy=3tdt$

$\frac{1}{y^2}dy-3tdt=0$

Yes variables are separable

(ii) $\frac{dy}{dt}=3y^2t$

Equation is not linear but can be linearized as a Bernoulli equation

2. Solve by separation of variable taking y and t to be positive. Check your answer by differentiation.

(i) $2tdy+2ydt=0$

Separating the variables:

Dividing both sides by "2ty"

$\frac{2tdy}{2ty}+\frac{2ydt}{2ty}=0$

$\frac{1}{y}dy+\frac{1}{t}dt=0$

Integrating both sides

$\int\frac{1}{y}dy+\int\frac{1}{t}dt=\int 0$

$\ln y+\ln t=c$

$\ln(yt)=c$

Taking anti log

$e^{ln(y,t)}=e^c$

$yt=e^c$

$y_{(t)}=At^{-1}$

$(e^c=A)$

Checking the answer

$2tdy+2ydt=0$

$2tdy=-2ydt$

$\frac{dy}{dt}=\frac{-2y}{2t}$

$\frac{dy}{dt}=\frac{-y}{t} \rightarrow (i)$

$y_{(t)}=At^{-1}$

$\frac{dy}{dt}=-At^{-2}$

Substituting into (i)

$-At^{-2}=-\frac{At^{-1}}{t}$

$\frac{-A}{t^2}=\frac{-A}{t^2}$

(b) $\frac{y}{y+t}dy+\frac{2t}{y+t}dt=0$

Multiplying both sides by (y + t)

$ydy+2tdt=0$

$\int ydy+\int 2tdt=\int 0$

$\frac{1}{2}y^2+t^2=c$

$\frac{1}{2}y^2=c-t^2$

$y^2=2(c-t^2)$

$y^2=2c-2t^2$

$(2c=A)$

$y=(A-2t^2)^{\frac{1}{2}}$

Validity:

$\frac{y}{y+t}dy+\frac{2t}{y+t}dt=0$

$ydy=-2tdt$

$\frac{dy}{dt}=\frac{-2t}{y} \quad \rightarrow (i)$

Taking derivative of answer

$y=(A-2t^2)^{\frac{1}{2}}$

$\frac{dy}{dt}=\frac{1}{2}(A-2t^2)^{\frac{-1}{2}}(-4t)$

$\frac{dy}{dt}=-2t(A-2t^2)^{\frac{-1}{2}}$

$\frac{dy}{dt}=\frac{-2t}{y}$

Substituting into (i)

$-2t(A-2t^2)^{\frac{-1}{2}}=\frac{-2t}{(A-2t^2)^{\frac{1}{2}}}$

$-2t(A-2t^2)^{\frac{-1}{2}}=-2t(A-2t^2)^{\frac{-1}{2}}$

3. Solve as a separable−variable equation and also as Bernoulli equation validity

$\frac{dy}{dt}=\frac{-t}{y}$

$ydy+tdt=0$

Integrating both sides

$\int ydy+\int tdt=\int 0$

$\frac{1}{2}y^2+\frac{1}{2}t^2=c$

$\frac{1}{2}y^2+\frac{1}{2}t^2=c$

$y^2=2c-t^2$

$y=(A-t^2)^{\frac{1}{2}}$

$(2c=A)$

The given equation is

$\frac{dy}{dt}=\frac{-t}{y}$ While the answer is

$y=(A-t^2)^{\frac{1}{2}}$

Taking derivative

$\frac{dy}{dt}=\frac{1}{2}(A-t^2)^{-\frac{1}{2}}(-2t)$

$\frac{dy}{dt}=-t(A-t^2)^{-\frac{1}{2}}$

Substituting values

$-t(A-t^2)^{\frac{-1}{2}}=\frac{-t}{(A-t^2)^{\frac{1}{2}}}$

$-t(A-t^2)^{\frac{-1}{2}}=-t(A-t^2)^{-\frac{1}{2}}$

Solving as a Bernoulli equation

$\frac{dy}{dt}=\frac{-t}{y}$

$\frac{dy}{dt}=-ty^{-1}$

$Let \quad m=-1$

$z=y^{1-m} \quad =y^2$

$R=0 \quad T=-t$

Substituting into following

$\frac{dz}{dt}+(1-m)Rz=(1-m)T$

$\frac{dz}{dt}+(1+1)(0)z=(1+1)(-t)$

$\frac{dz}{dt}=-2t \qquad u=0$

$w=-2t$

$z_{(t)}=e^{-\int udt}(A+\int we^{\int udt}dt)$

$z_{(t)}=e^{-c}(A-2\int te^{c}dt)$

$z_{(t)}=e^{-c}(A-2e^{c}\int t\,dt)$

$z_{(t)}=e^{-c}(A-2e^{c}\frac{t^2}{2})$

$z_{(t)}=Ae^{-c}-t^2 \qquad (Ae^{-c}=A)$

$y^2=A-t^2 \qquad (z_{(t)}=y^2)$

$y=(A-t^2)^{\frac{1}{2}}$

Q.4 Solve $\frac{dy}{dt}=3y^2t$

$\frac{1}{y^2}dy=3tdt$

$y^{-2}dy=3tdt$

$y^{-2}dy-3tdt=0$

Integrating both sides W.r.t "t"

$\int y^{-2}dy-\int 3tdt=\int 0$

$-y^{-1}-1.5t^2=c$

$-y^{-1}=c+1.5t^2 \qquad (-c=c)$

$y=(c-1.5t^2)^{-1}$

Solving as a Bernoulli equation

$\frac{dy}{dt}=3y^2t$

$m=2,$

$z=y^{1-m}=y^{1-2}=y^{-1},$

$R=0\,, \qquad T=3t$

$\frac{dz}{dt}+(1-m)Rz=(1-m)T$

$\frac{dz}{dt}+(1-2)(0)z=(1-2)3t$

$\frac{dz}{dt}=-3t \qquad u=0$

$w=-3t$

Applying formula for general solution

$z_{(t)}=e^{-\int udt}(A+\int we^{\int udt}dt)$

$z_{(t)}=e^{-c}(A-3\int te^{c}dt)$

$z_{(t)}=e^{-c}(A-3e^{c}\int tdt)$

$z_{(t)}=e^{-c}(A-1.5e^{c}t^2)$

$z_{(t)}=Ae^{-c}-1.5t^2$

$As \quad z=y^{-1}$

$y^{-1}=(A-1.5t^2) \qquad (Ae^{-c}=A)$

$y=(A-1.5t^2)$

Q.5 Verify the correctness of intermediate solution $z_{(t)}=At^2+2t$ in example 4 by show that its derivative $\frac{dz}{dt}$ is consistent with the linearized differential equation.

$z_{(t)}=At^2+2t \quad \rightarrow(i)$

$\frac{dz}{dt}=2At+2 \quad \rightarrow(ii)$

Given equation (differential equation) is

$\frac{dz}{dt}-\frac{2}{t}2=-2$

Substituting from (i) and (ii)

$2At+2-\frac{-2}{t}(At^2+2t)=-2$

$2At+2-2At-4=-2$

$-4+2=-2$

$-4+2-2=0$

$0=0$

6: The Qualitative Graphic Approach:

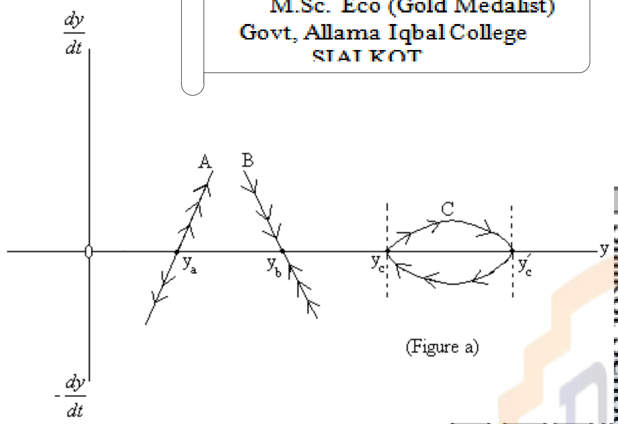
At times we may not be able to find a quantitative solution from a given differential equation. Yet, in such cases it may nonetheless be possible to ascertain the qualitative properties of the time path−primarily, whether $y_{(t)}$ converges by directly observing the differential equation itself or by analyzing its graph. Even when quantitative solution is available, we may still employ the techniques of qualitative analysis if the qualitative aspect of the time path is our principal or exclusive concern.

The Phase Diagram:

Given a first−order differential equation In the general form

$\frac{dy}{dt}=f(y)$

Either linear or nonlinear in the variable y, we can plot $\frac{dy}{dt}$ against y as in figure a. Such a geometric representation feasible whenever $\frac{dy}{dt}$ is a function of y alone, is called a <u>phase diagram</u> and graph representing the function f, a <u>phase line</u>



(Figure a)

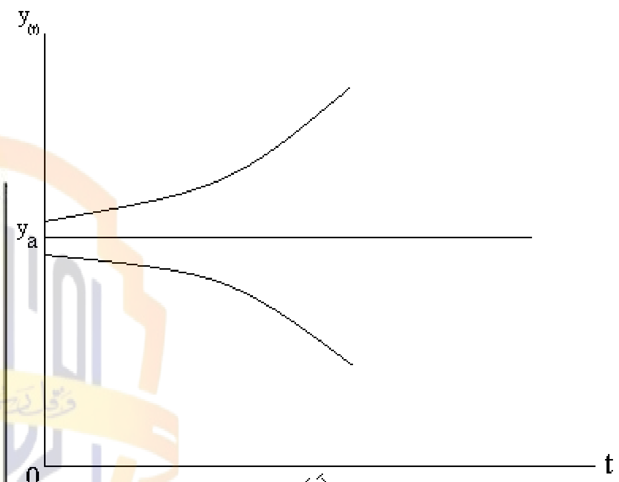
Once a phase line is known, its configuration will impart significant qualitative information regarding the time path $y_{(t)}$ the clue to this lies in following two general remarks.

(i) Anywhere above the horizontal axis where $\left(\frac{dy}{dt}>0\right)$ y must be increasing overtime as for as the y axis is concerned, must be moving from left to right. By analogous reasoning, any point below the horizontal axis must be associated with a leftward movement in the variable y, because the negativity of $\frac{dy}{dt}$ means that y decreases overtime

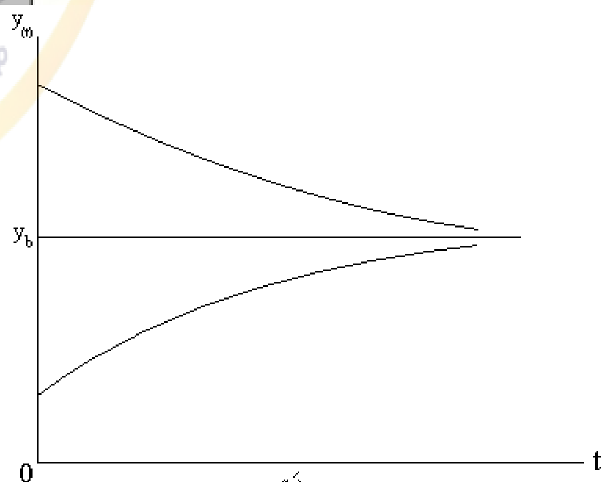
(ii) An equilibrium level of y−in the inter temporal sense of term if it exists, can occur only on the horizontal axis, where $\frac{dy}{dt}=0$ (y stationary over time)

<u>Types of Time Path:</u>

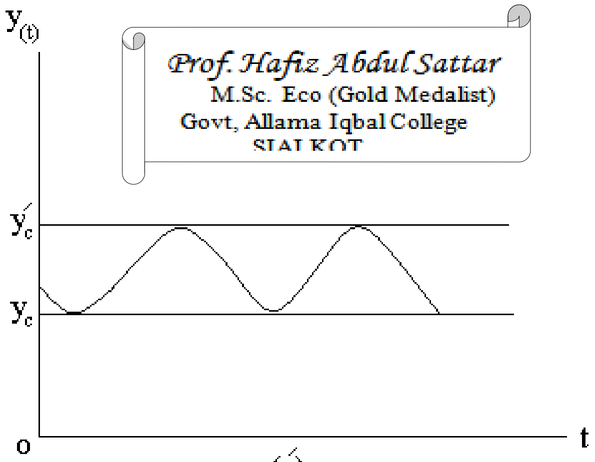
Phase line A has equilibrium at point y_a but above as well as below that point the arrowheads consistently lead away from equilibrium. Thus although equilibrium can be attained if it happens that $y_{(0)}=y_a$ the more usual case of $y_0 \neq y_a$ will result in y being over increasing [if $y_{(0)} > y_a$] or ever decreasing [if $y_{(0)} < y_a$]. The time path $y_{(t)}$ implied by the phase line A can therefore be represented by the curve shown in following figure $(b)\ a'$



(a')



(b')



(c')

In contrast, phase line b implies a stable equilibrium at $y_{(0)}$ if $y_{(0)} = y_b$ equilibrium prevails at once. But the important feature of phase line B is that, even if $y_{(0)} \neq y_b$ the movement along the phase line B will guide y towards the level of y_b. The time path $y_{(t)}$ corresponding to this type of phase line should therefore be of the form shown in figure $b(b')$, which is reminiscent of the dynamic market model.

A (finite) positive slope such as at point y_a make for dynamic instability whereas (finite) negative slope such as at y_b implies dynamic stability.

Mathematically

If $\frac{d}{dy}(\frac{dy}{dt}) < o$ the equilibrium is stable

If $\frac{d}{dy}(\frac{dy}{dt}) > o$ the equilibrium is unstable

This generalization can help us to draw qualitative inferences about given differential equation without even plotting their phase lines. Taking a linear differential equation

$$\frac{dy}{dt} + ay = b \quad Or \quad \frac{dy}{dt} = -ay + b$$

Since the phase line will obviously have the (constant) slope –a, here assumed non−zero, we may immediately infer that

$$a \Leftrightarrow y_{(t)} \begin{Bmatrix} converges\ to \\ diverges\ from \end{Bmatrix} equilibrium$$

This result coincides perfectly with, what the quantitative solution of this equation tells us:

$$y_{(t)} = \left[y_{(0)} - \frac{b}{a} \right] e^{-at} + \frac{b}{a}$$

Starting from a non−equilibrium point, the convergence of $y_{(t)}$ hinges on the prospect that $e^{-at} \to 0 \quad as \quad t \to \infty$ this can happen if and only if a>0; if a<0 then $e^{-at} \to \infty \quad as \quad t \to \infty \quad and \quad y_{(t)}$ cannot converge. Thus our conclusion is one and the same, whether it is arrived at quantitatively or qualitatively

It remains to discuss phase line c, which being a closed loop sitting across the horizontal axis does not qualify as a function but show instead a relation between $\frac{dy}{dt}$ and y. the interesting new element that emerges in this case is the possibility of a periodically fluctuating time path.

Exercise 14.6:

Q.1 Plot the phase line for each of following and discuss its qualitative implications.

(a) $\frac{dy}{dt} = y - 7$

$put\ it\ equal\ to\ zero$

$$\frac{dy}{dt} = y - 7 = 0$$

$$y = 7$$

$$\frac{dy}{dt} = y - 7 = 0$$

$$\frac{dy}{dt} = 3 - 7 = -4$$

$$\frac{dy}{dt} = 5 - 7 = -2$$

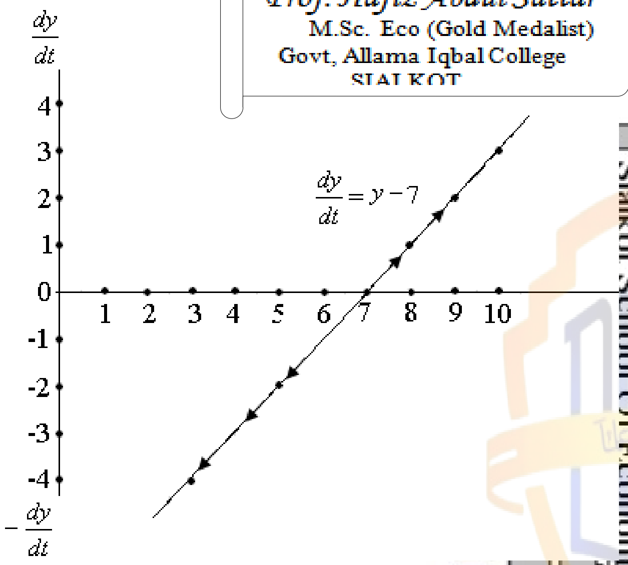
$$\frac{dy}{dt} = 7 - 7 = 0$$

$$\frac{dy}{dt} = 8 - 7 = 1$$

$$\frac{dy}{dt} = 9 - 7 = 2$$

$$\frac{dy}{dt}=10-7=3$$

$\frac{dy}{dt}$	y
-4	3
-2	5
0	7
1	8
2	9
3	10



Upward sloping phase line dynamically unstable equilibrium

(b) $\frac{dy}{dt}=1-5y$

put it equal to zero

$5y=1$

$y=\frac{1}{5}=0.2$

$$\frac{dy}{dt}=1-5(-0.4)=3$$

$$\frac{dy}{dt}=1-5(-0.2)=2$$

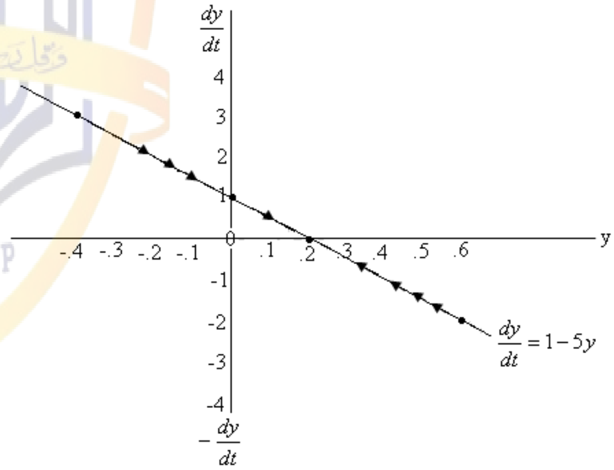
$$\frac{dy}{dt}=1-5(0)=1$$

$$\frac{dy}{dt}=1-5(0.2)=0$$

$$\frac{dy}{dt}=1-5(0.4)=-1$$

$$\frac{dy}{dt}=1-5(0.6)=-2$$

$\frac{dy}{dt}$	y
3	-0.4
2	-0.2
1	0
0	0.2
-1	0.4
-2	0.6



Downward sloping phase line depicts dynamically stable equilibrium

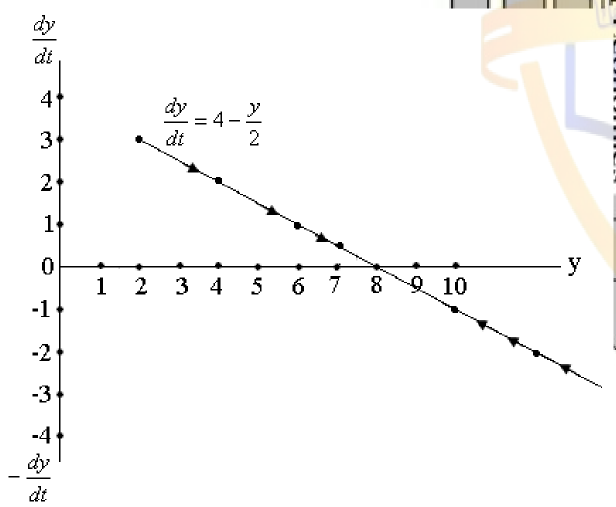
(c) $\frac{dy}{dt}=4-\frac{y}{2}$

put it equal to zero

$4-\frac{y}{2}=0\Rightarrow\frac{y}{2}=4\Rightarrow y=8$

$$\frac{dy}{dt}=4-\frac{2}{2}=3$$
$$\frac{dy}{dt}=4-\frac{4}{2}=2$$
$$\frac{dy}{dt}=4-\frac{6}{2}=1$$
$$\frac{dy}{dt}=4-\frac{8}{2}=0$$
$$\frac{dy}{dt}=4-\frac{10}{2}=-1$$
$$\frac{dy}{dt}=4-\frac{12}{2}=-2$$

$\frac{dy}{dt}$	y
3	2
2	4
1	6
0	8
-1	10
-2	12



Downward sloping phase line depicts dynamically stable equilibrium

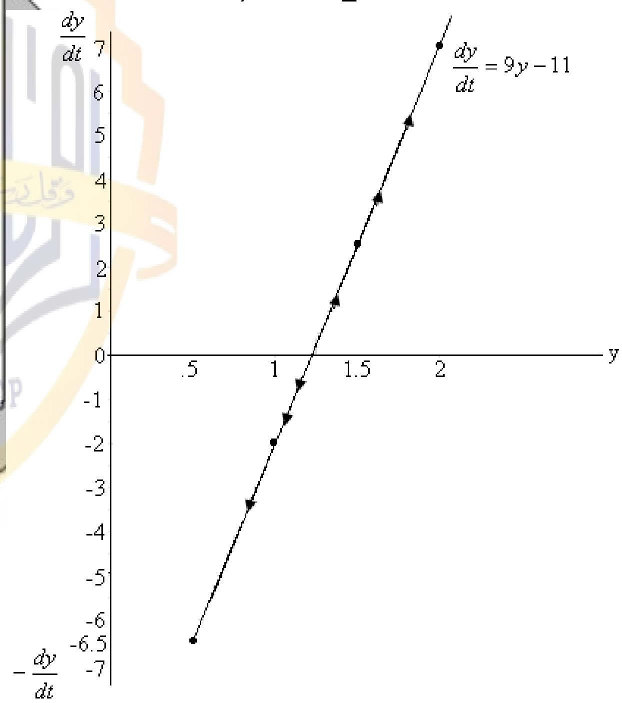
(d) $\frac{dy}{dt}=9y-11$

Put it equal to zero

$$y=\frac{11}{9}=1.2$$

$$\frac{dy}{dt}=9(0.5)-11=-6.5$$
$$\frac{dy}{dt}=9(1)-11=-2$$
$$\frac{dy}{dt}=9(1.2)-11=0$$
$$\frac{dy}{dt}=9(1.5)-11=2.5$$
$$\frac{dy}{dt}=9(2)-11=7$$

$\frac{dy}{dt}$	y
-6.5	0.5
-2	1
0	1.2
2.5	1.5
7	2



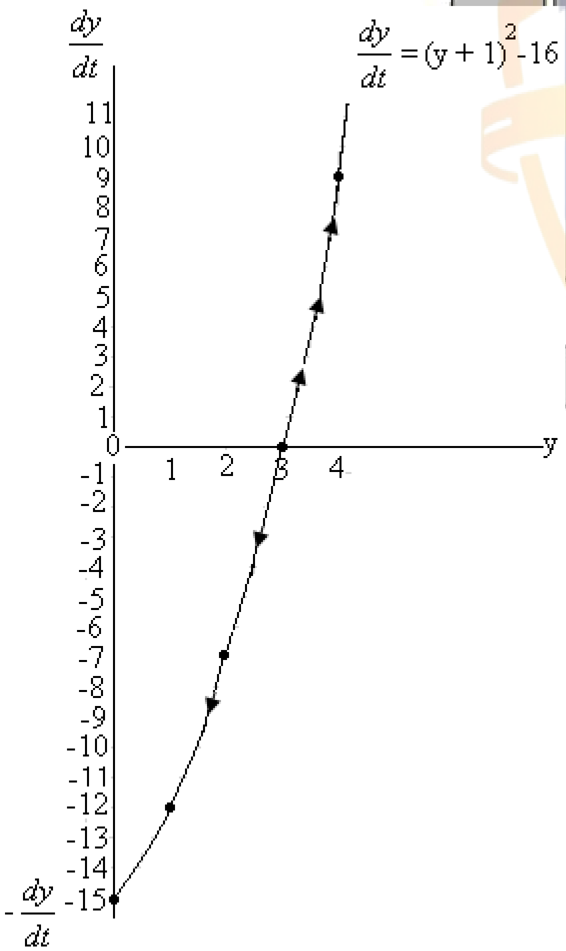
Upward sloping phase line depicts dynamically unstable equilibrium

Q.2. Plot the phase line for each of the following and interpret.

(a) $\frac{dy}{dt}=(y+1)^2-16 \quad (y\geq 0)$

$$\frac{dy}{dt}=(0+1)^2-16=-15$$
$$\frac{dy}{dt}=(1+1)^2-16=-12$$
$$\frac{dy}{dt}=(2+1)^2-16=-7$$
$$\frac{dy}{dt}=1(3+1)^2-16=-7$$
$$\frac{dy}{dt}=(3+1)^2-16=0$$
$$\frac{dy}{dt}=(4+1)^2-16=9$$

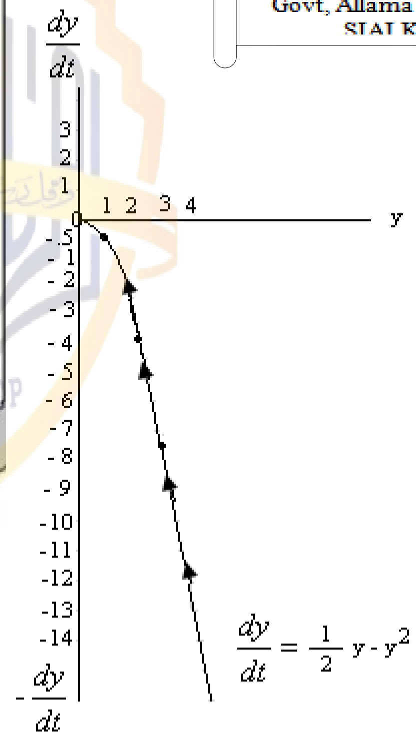
y	$\frac{dy}{dt}$
0	-15
1	-12
2	-7
3	0
4	9



Upward sloping phase line depicts dynamically unstable equilibrium

(b) $\frac{dy}{dt}=\frac{1}{2}y-y^2 \quad (y\geq 0)$

$$\frac{dy}{dt}=\frac{1}{2}(0)-(0)^2=0$$
$$\frac{dy}{dt}=\frac{1}{2}y-y^2$$
$$\frac{dy}{dt}=\frac{1}{2}(1)-(1)^2=0.5$$
$$\frac{dy}{dt}=\frac{1}{2}(2)-(2)^2=-4$$
$$\frac{dy}{dt}=\frac{1}{2}(3)-(3)^2=-7.5$$
$$\frac{dy}{dt}=\frac{1}{2}(4)-(4)^2=-14$$



Downward sloping phase line depicts dynamically stable equilibrium

Q.3. Given $\frac{dy}{dt}=(y-3)(y-5)=y^2-8y+15$

Deduce that there are two possible equilibrium level of y when y=3 and the other at y=5

$$\frac{dy}{dt}=y^2-8y+15 \quad when \quad y=3$$

$$\frac{dy}{dt}=(3)^2-8(3)+15=9-24+15=0$$

$when \quad y=5$

$$\frac{dy}{dt}=(5)^2-8(5)+15=25-40+15=0$$

Hence there are two possible equilibrium level of y because

$$\frac{dy}{dt}=0 \quad at \quad y=3 \quad and \quad y=5$$

(b) Find the sign of

$\frac{d}{dy}\left(\frac{dy}{dt}\right) at \quad y=3 \quad and \quad y=5$ what can you infer from these

$$\frac{dy}{dt}=y^2-8y+15$$

$$\frac{d}{dt}\left(\frac{dy}{dt}\right)=2y-8 \quad \rightarrow (i)$$

Put y=3 in (i)

$$\frac{d\left(\frac{dy}{dt}\right)}{dy}>0$$

Hence unstable equilibrium at y=5

7: Solow Growth Model: The Framework:

In the Domar model, output is explicitly stated as a function of capital alone.

$$k=\rho \; K$$

The absence of the labor input in the production function carries the implication that labor is always combined with capital in a fixed proportion. Solow in contrast, seeks to analyze the case where capital and labor can be combined in varying proportion.

(i) The production function appears in the form

$$Q=f(K,L) \rightarrow (i) \qquad (K, L>0)$$

(a) It is assumed that f_K and f_L are positive (positive marginal product)

(b) f_{KK} and f_{LL} are negative (diminishing return to each input)

(ii) The production function (f) is taken to be linearly homogenous (constant return to scale). Consequently it is possible to write

$$Q=Lf\left(\frac{K}{L},1\right)=L\varphi(k) \quad where \quad k=\frac{K}{L}$$

(iii) $K^0\left(\equiv\frac{dk}{dt}\right)=sQ$ [Constant proportion of Q is invested]

The symbol s represents a (constant) marginal propensity to save

(iv) $\frac{L^0}{L}\left(¿\frac{dL/dt}{L}\right)=\lambda \quad (\lambda>0)$

[Labor force grows exponentially]

The symbol λ represents a (constant) rate of growth of labor.

From (i)

$$Q=f(L, K)$$

$$Q=Lf\left(\frac{K}{L},1\right)$$

$$Q=L\varphi(k) \qquad \left(k=\frac{K}{L}\right)$$

$$\overset{0}{K}=sQ \rightarrow (i)'$$

$$\overset{0}{K}=s(L\varphi(k))$$

$$\overset{0}{K}=sL\varphi(k) \rightarrow (ii)$$

$$As \quad k=\frac{K}{L} \quad then \quad K=Lk$$

$$\overset{0}{K}=Lk^0+kL^0 \rightarrow (iii)$$

Substituting (iii) in (i)

$$Lk^0+kL^0=sL\varphi(k)$$

$$Lk^0+k(\lambda L)=sL\varphi(k)$$

Eliminating the common factor L

$$k^0+k\lambda=s\varphi(k)$$

$$k^0=s\varphi(k)-\lambda k$$

$$k^0=s\varphi(k)-\lambda k$$

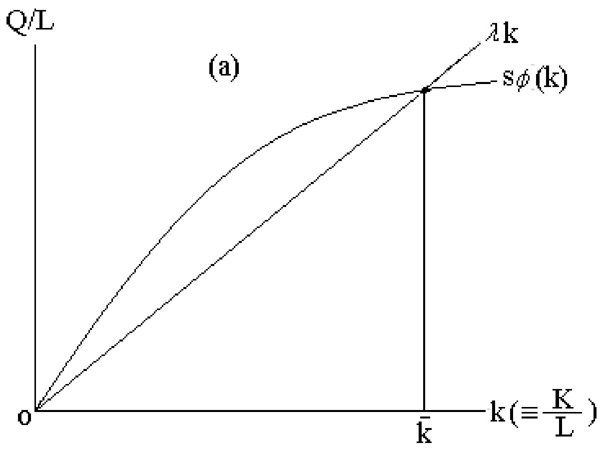
This equation is the fundamental equation of Solow growth model.

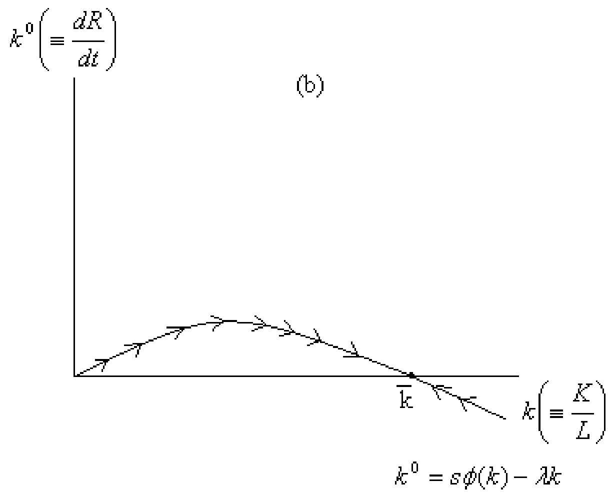
A Qualitative Graphic Analysis:

The fundamental equation of Solow growth model can be analyzed qualitatively. To this end we should plot a phase line with k^0 on vertical axis and k on the horizontal.

The equation contains two terms on the right. The λk term a linear function of k will obviously show up as a straight line with a zero vertical intercept and slope equal to λ. The $s\varphi(k)$ term on the other hand plots a curve that increases at decreasing rate, like $\varphi(k)$, since $s\varphi k$ is merely a constant fraction of the $\varphi(k)$ curve if we consider k to be indispensible factor of production we must state the $\varphi(k)$ curve from the point of origin: this is because if k=0 and thus k=0, Q must also be zero, as will be $\varphi(k)$ and $s\varphi(k)$.

Based upon these two curves, the value of k^0 for each value of k can be measured by vertical distance between the two curves. Plotting the value of k^0 against k will then yield the phase line we need. Since the two curves in diagram a interest when the





Capital−labor ratio is $\bar{k}$, the phase line in diagram b must cross the horizontal axis at $\bar{k}$. This marks the $\bar{k}$ as the (inter temporal) equilibrium capital−labor ratio.

Inasmuch as the phase line has a negative slope at $\bar{k}$, the equilibrium is readily identified as a stable one; given any (positive) initial value of k, the dynamic movement of the model must lead us convergent to the equilibrium level $\bar{k}$. The significant point is that once this equilibrium is attained –and thus the capital labor ratio is (by definition) unvarying over time−capital must thereafter grow apace with labor, at the identical rate λ.

Such a situation in which the relevant variables all grow at an identical rate, is called a steady state−a generalization of the concept of stationary state.

A Quantitative illustration:

There is no clear understanding of the model because of the presence of a general function $\varphi(k)$ in the model. But if we specify the production function to be a linearly homogenous Cob Douglas function, for instance, then a quantitative solution can be found as well. Let us write production function as

$$Q=K^{\alpha}L^{\beta} \qquad \alpha+\beta=1$$
$$Q=K^{\alpha}L^{1-\alpha}$$
$$Q=K^{\alpha}\frac{L}{L^{\alpha}}$$
$$Q=\left(\frac{K}{L}\right)^{\alpha}.\ L$$
$$Q=L\left(\frac{K}{L}\right)^{\alpha}$$
$$Q=Lk^{\alpha}$$
$$\frac{Q}{L}=k^{\alpha}$$
$As \quad Q=L\varphi(k)$
$$\frac{Q}{L}=\varphi(k)$$
$then \quad \varphi(k)=k^{\alpha}$

Substituting k^{α} in place of $\varphi(k)$ in following equation
$$k^0=s\varphi(k)-\lambda k$$
$$k^0=sk^{\alpha}-\lambda k$$
$$k^0+\lambda k=sk^{\alpha}$$
$$\frac{dk}{dt}+\lambda k=sk^{\alpha} \qquad k^0=\frac{dk}{dt}$$
This is the Bernoulli equation in variable k.

$With \quad m=\alpha \quad let \quad z=k^{1-m}=k^{1-\alpha}$

$R=\lambda \quad T=s$

Applying formula to linear the Bernoulli equation
$$\frac{dz}{dt}+(1-m)Rz=(1-m)T$$
$$\frac{dz}{dt}+(1-\alpha)\lambda z=(1-\alpha)s$$
$$\frac{dz}{dt}+\underbrace{(1-\alpha)\lambda}_{a}z=\underbrace{(1-\alpha)s}_{b}$$
$$z_{(t)}=Ae^{-at}+\frac{b}{a}$$
$$z_{(t)}=Ae^{-(1-\alpha)\lambda t}+\frac{(1-\alpha)s}{(1-\alpha)\lambda}$$
$$z_{(t)}=Ae^{-(1-\alpha)\lambda t}+\frac{s}{\lambda}$$
$$z_{(0)}=A+\frac{s}{\lambda}$$
$$z_{(0)}-\frac{s}{\lambda}=A$$
$$z_{(t)}=\left[z_{(0)}-\frac{s}{\lambda}\right]e^{-(1-\alpha)\lambda t}+\frac{s}{\lambda}$$
$$k^{1-\alpha}=\left[k_{(0)}^{1-\alpha}-\frac{s}{\lambda}\right]e^{-(1-\alpha)\lambda t}+\frac{s}{\lambda}$$
As $t\rightarrow 0$ the first term on the right hand side will disappears

$Note \quad k^0=sQ$
$$\frac{k^0}{Q}=s$$
$\frac{1}{Q}=s \quad s \ shows \ capital \ output \ ratio.\ \lambda \ is \ growth \ rate \ of \ labour$

$$k^{1-\alpha}\rightarrow\frac{s}{\lambda}$$
$$k^{1-\alpha}\rightarrow\left(\frac{s}{\lambda}\right)^{\frac{1}{1-\alpha}}$$
$$\frac{K}{L}\rightarrow\left(\frac{s}{\lambda}\right)^{\frac{1}{1-\alpha}}$$
$Note \quad k^0=sQ$
$$\frac{k^0}{Q}=s$$
$\frac{1}{Q}=s \quad s \ shows \ capital \ output \ ratio.\ \lambda \ is \ growth \ rate \ of \ labour$

Therefore the capital−labor ratio will approach a constant as its equilibrium value. This equilibrium or steady state value $\left(\frac{s}{\lambda}\right)^{\frac{1}{1-\alpha}}$ varies directly with propensity to save and inversely with the rate of growth of labor λ.

Exercise 14.7:

Q.1. Divide $k^0=s\varphi(k)-\lambda k$ through by k and interpret the resulting equation in term of the growth rate of k, K and L.

Answer:

$$k^0=s\varphi(k)-\lambda k$$
Dividing through by k

$\frac{k^0}{k}=\frac{s\varphi(k)}{k}-\frac{\lambda k}{k} \quad \rightarrow (i)$

As $K^0=sQ=sL\varphi(k)$

Therefore $k=\frac{K^0}{L}s\varphi(k)$

$$k=\frac{K}{L} \;\; and \;\; \lambda=\frac{L^0}{L}$$

Moreover

Substituting this information in equation(i)

$\frac{k^0}{k}=\frac{K^0/L}{K/L}-\lambda$

$\frac{k^0}{k}=\frac{K^0}{K}-\frac{L^0}{L} \quad \rightarrow (ii)$

Where, $\frac{k^0}{k}$ is the rate of growth of capital labor ratio $\frac{K^0}{K}$ is rate of growth of capital stock and $\frac{L^0}{L}$ is the rate of growth of labor force. The equation (ii) shows that

1. If capital stock and labor force grow at a same rate then the growth rate of capital−labor ratio will be zero.
2. If the growth rate of capital stock exceeds the growth rate of labor force, growth rate of capital−labor ratio will be positive.
3. If the growth rate of capital stock lags behind the growth rate of labor force, growth rate of capital− labor ratio will be negative.

Q.2 Show that, if capital is growing at the rate λ $(i.e. \quad K=Ae^{\lambda t})$, net investment I must also be growing at the rate λ .

$As \quad K=Ae^{\lambda t} \quad \rightarrow (i)$

$Then \quad K^0=\lambda Ae^{\lambda t} \rightarrow (ii)$

Finding the growth rate of capital

$\frac{K^0}{K}=\frac{\lambda Ae^{\lambda t}}{Ae^{\lambda t}}=\lambda$

Since $k^0=I$

Then $I=\lambda Ae^{\lambda t}$

$I^0=\lambda^2 Ae^{\lambda t}$

$\frac{I^0}{I}=\frac{\lambda^2 Ae^{\lambda t}}{\lambda Ae^{\lambda t}}=\lambda$

Hence both capital and net investment are growing at the same rate λ

Q.4 The original input variable of Solow model is K and L but the fundamental equation focuses on the capital labor ratio k instead. What assumption in the model is responsible for this shift of focus? Explain?

Answer:

The assumption that capital and labor can be combined in varying proportion in production process is responsible for this shift of focus. It means production isoquant will be convex to the origin lather than L−shaped

Q.5 Draw a phase diagram for each of the following and discuss the qualitative aspects of the time path $y_{(t)}$.

(a) $y^0=3-y-\ln y$

$y^0=3-(1)-\ln(1)=2$

$y^0=3-(1.5)-\ln(1.5)=1.1$

$y^0=3-(2)-\ln(2)=0.31$

$y^0=3-(2.5)-\ln(2.5)=-0.42$

$y^0=3-(3)-\ln(3)=-1.0$

Hence

Y	Y^0
1	2
1.5	1.1
2	0.31
2.5	−0.42
3	−1.0

(b) $y^0=e^y-(y+2)$

$y^0=e^{-2}-(-2+2)=0.14$

$y^0=e^{-1}-(-1+2)=-0.63$

$y^0=e^0-(0+2)=-1$

$y^0 = e^1 - (1+2) = -0.28$

$y^0 = e^2 - (2+2) = 3.39$

Hence

Y	Y^0
-2	0.14
-1	-0.63
0	-1
1	-0.28
2	3.39